MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

Project **Mansourah Massarah Gold Project**
Customer: **MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: **CALCULATION**
3840 FLOCCULANT CONCRETE
DESIGN CALCULATION

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TABLE OF CONTENTS

DEFINITIONS AND ABBREVIATIONS ……3

1. INTENT……...4

2. EXECUTIVE SUMMARY……4

3. SCOPE……4

4. DESIGN BASIS……5
   - 4.1 Materials……6
   - 4.2 Loading……6
   - 4.3 Load Combinations……7
   - 4.4 Minimum Reinforcing Steel (Rebar) Size……10
   - 4.5 Minimum Concrete Cover……10
   - 4.6 Allowable Soil Bearing Capacity……10

5. DESIGN CALCULATION……11
   - 5.1 Design calculation of Footing F1……12
   - 5.2 Design calculation of Column C1……23
   - 5.3 Design calculation of Footing F2……26
   - 5.4 Design calculation of Column C2……37

References……40

ATTACHMENT 1 OTE INPUT DRAWING & LOAD DATA DRAWING……41

ATTACHMENT 2 COMMENT RESOLUTION LOG SHEET…… …… ..46

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DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
EMPLOYER	:	'EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	:	Ma'aden Project Management Team
'MMGP Project'	:	Mansourah Massarah Gold Processing Project
Attribute	:	Information element that describes a document or tag.
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
Sub-Supplier	:	A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor.
QA	:	Quality Assurance
QC	:	Quality Control
Works	:	Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities

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1. INTENT

This document is to prepare design of foundation for Flocculant Re-agent Building (3840). The analysis & design is carried out in accordance with project specifications and codes & standards. The design calculation ensures the safety and serviceability of the foundation system.

2. EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.

MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.

The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week

3. SCOPE

In the specified area, there are 4 nos. Flocculant Reagent Building. The analysis and design have been carried out in excel, considering the load data attached here with in Attachment 1. The load combinations for serviceability check and member's strength calculation are as per SBC.

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4. DESIGN BASIS

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4.1 Material

A. Concrete Strength of 28 days

Foundation and Structural Members: 35 MPa

Blinding Concrete below foundation bases: 15 MPa

B. Reinforcement Grade

Deformed Bar : ASTM A 615 / A 615M Grade 60 (fy = 420MPa)

Welded Wire Fabric : ASTM A 185(Plain) / A 497(Deformed)

C. Grouting

Grout shall be 50 mm cementitious of Strength 40 MPa.

4.2 Loading

A. Primary Load

1 D = Dead load

2 Cl = Collateral Load

3 Lr = Roof Live

4 Cr = Crane Load

5 WL1 = Wind load from left to right with internal pressure

6 WL2 = Wind load from left to right with internal suction

7 WR1 = Wind load from right to left with internal pressure

8 WR2 = Wind load from right to left with internal suction

9 WL1 = Wind load parallel to ridge from end wall with internal pressure

10 WL2 = Wind load parallel to ridge from end wall with internal suction

11 WL-1 = Wind load parallel to ridge from end wall with internal pressure

12 WL-2 = Wind load parallel to ridge from end wall with internal suction

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13 SL = Seismic load from left to right

14 SR = Seismic load from right to left

15 T1 = Temperature load for 25 degree

16 T2 = Temperature load for -25 degree

4.3 Load combination

A. The following load combination are considered for base pressure, sliding & overturning check.

101 D+Cl

102 D+Cl+Lr+Cr

103 D+Cl+Cr+T1

104 D+Cl+Cr+T2

105 D+Cl+Cr+0.75Lr+0.75T1

106 D+Cl+Cr+0.75Lr+0.75T2

107 D+Cl+WL1

108 D+Cl+WL2

109 D+Cl+WR1

110 D+Cl+WR2

111 D+Cl+WLn1

112 D+Cl+W_Ln2

113 D+Cl+W_Ln-1

114 D+Cl+W_Ln-2

115 D+Cl+0.7SL

116 D+Cl+0.7SR

117 D+0.75Lr+0.75WL1

118 D+0.75Lr+0.75WL2

119 D+0.75Lr+0.75WR1

120 D+0.75Lr+0.75WR2

121 D+0.75Lr+0.75W_Ln1

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122 D+0.75Lr+0.75W_Ln2

123 D+0.75Lr+0.75W_Ln-1

124 D+0.75Lr+0.75W_Ln-2

125 D+0.75Lr+0.525SL

126 D+0.75Lr+0.525SR

127 0.6D+WL1

128 0.6D+WL2

129 0.6D+WR1

130 0.6D+WR2

131 0.6D+W_Ln1

132 0.6D+W_Ln2

133 0.6D+W_Ln-1

134 0.6D+W_Ln-2

135 0.6D+0.7SL

136 0.6D+0.7SR

B) The following load combination are considered for member strength calculation.

201 1.4(D+Cl)

202 1.4(D+Cl+Cr)

203 1.4(D+Cl+T1)+0.5Lr

204 1.4(D+Cl+T2)+0.5Lr

205 1.2(D+Cl)+1.6Lr+0.8WL1

206 1.2(D+Cl)+1.6Lr+0.8WL2

207 1.2(D+Cl)+1.6Lr+0.8WR1

208 1.2(D+Cl)+1.6Lr+0.8WR2

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209 1.2(D+Cl)+1.6Lr+0.8W_Ln1
210 1.2(D+Cl)+1.6Lr+0.8W_Ln2
211 1.2(D+Cl)+1.6Lr+0.8W_Ln-1
212 1.2(D+Cl)+1.6Lr+0.8W_Ln-2
213 1.2(D+Cl)+0.5Lr+1.6WL1
214 1.2(D+Cl)+0.5Lr+1.6WR1
215 1.2(D+Cl)+0.5Lr+1.6WL2
216 1.2(D+Cl)+0.5Lr+1.6WR2
217 1.2(D+Cl)+0.5Lr+1.6W_Ln1
218 1.2(D+Cl)+0.5Lr+1.6W_Ln2
219 1.2(D+Cl)+0.5Lr+1.6W_Ln-1
220 1.2(D+Cl)+0.5Lr+1.6W_Ln-2
221 1.2D+SL
222 1.2D+SR
223 0.9D+1.6WL1
224 0.9D+1.6WL2
225 0.9D+1.6WR1
226 0.9D+1.6WR2
227 0.9D+1.6W_Ln1
228 0.9D+1.6W_Ln2
229 0.9D+1.6W_Ln-1
230 0.9D+1.6W_Ln-2
231 0.9D+1.0SL
232 0.9D+1.0SR

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4.4 Minimum Reinforcing Steel (Rebar) Size

The minimum rebar size for use as tension, compression reinforcement shall be 12 mm except where welded wire fabric is used for ties and spirals the minimum rebar size shall be 10 mm.

4.5 Minimum Clear Cover

	Member	Minimum Cover(mm)
Exposed to weather or contact with ground	Foundation & Columns	50

4.6 Allowable Soil Bearing Capacity

The net safe bearing capacity of soil for foundations design has been considered as 400 kN/m^2 at 1.5 m depth below EGL (i.e. RL 911.0 m) refer Table 5 - BH-7, BH-12 & BH-26 of Geotechnical Report (MD-216-3000-EG-CV-RPT-1008).

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5. DESIGN CALCULATION

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5.1 Design calculation of Footing F1

FOUNDATION MKD. F1

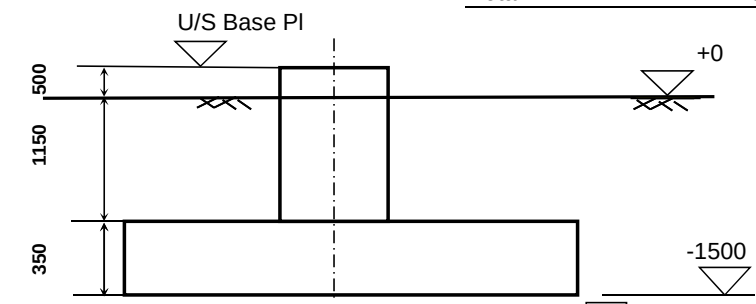
	Along-X	Along-Z
Size of column :	0.50m	0.50m
Size of fdn. raft :	2.00m	2.00m
Thickness of raft :	0.35 m	
Original ground level:	0.00 m	
Finished floor level :	0.00 m	
U/S level of foundation :	-1.50 m	
U/S level Base Plate :	0.50 m	
Unit weight of conc :	24 kN/m^3	
Unit weight of soil :	18 kN/m^3	

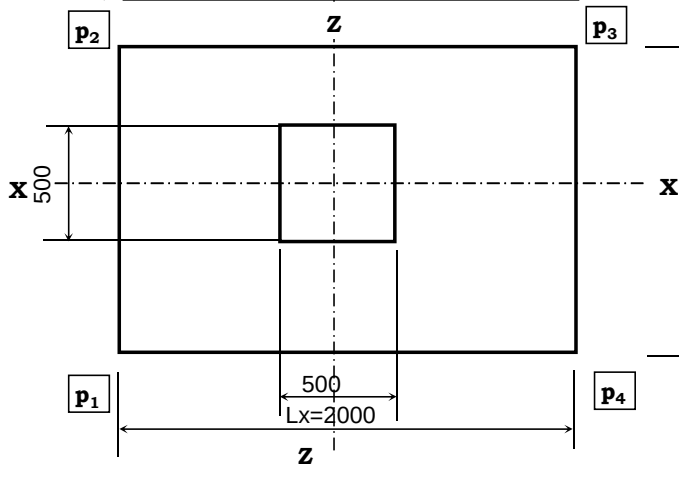
Calculation of Weight of Foundation :

Weight of raft & Col.= (2*2*0.35*24)+(0.5*0.5*1.65*24) = 43.50KN

Weight of soil = ((2*2)-(0.5*0.5))*1.15*18 = 77.63KN

Total = 121.13KN





Design Input:

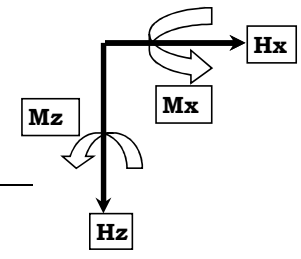
Characteristic strength of concrete, f '$_c$=	35 MPa
Yield stress of steel reinforcement, f_y=	420 MPa
Clear cover for main reinforcement bars =	50mm
Dia. of bars provd. along X direction at bottom =	12mm
Dia. of bars provided along Z direction at bottom=	12mm
Mininum percentage of reinforcement at bottom=	0.18%

Effective depth of slab along X direction = 294mm = 0.294m

Effective depth of slab along Z direction = 282mm = 0.282m

Clear projection along Z direction = 0.75m

Clear projection along X direction = 0.75m



H_X=Force along X-X direction

Hz=Force along Z-Z direction

M_X=Moment about X-X direction

Mz=Moment about Z-Z direction

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FOUNDATION MKD. F1

(Refer attached load data dwg.)

Sl. No.	Basic Load Case	Main Frame Reaction		Portal bracing reaction		
		V (kN)	Hx (kN)	Mz (kNm)	Hz (kN)	Mx (kNm)
1	Dead Load (D)	8.22	0.40	0.0	0.0	0.0
2	Collateral Load (Cl)	10.36	1.78	0.0	0.0	0.0
3	Roof Live (Lr)	5.17	0.44	0.0	0.0	0.0
4	Crane Load (Cr)	-1.05	-0.16	0.0	0.0	0.0
5	Wind_Left1 (WL1)	-24.87	-15.26	0.0	0.0	0.0
6	Wind_Left2 (WL2)	-19.46	-19.21	0.0	0.0	0.0
7	Wind_Right1 (WR1)	-3.51	11.99	0.0	0.0	0.0
8	Wind_Right2 (WR2)	1.97	8.08	0.0	0.0	0.0
9	Wind_Long1 (W_Ln1)	-48.13	5.89	0.0	-20.0	0.0
10	Wind_Long2 (W_Ln2)	-42.66	1.96	0.0	-20.0	0.0
11	Wind_Long-1 (W_Ln-1)	-44.37	5.92	0.0	20.0	0.0
12	Wind_Long-2 (W_Ln-2)	-38.91	2.00	0.0	20.0	0.0
13	Seismic_L (SL)	-8.04	-0.40	0.0	10.0	0.0
14	Seismic_R (SR)	8.04	0.40	0.0	-10.0	0.0
15	Temparature 25 degree (T1)	0.45	0.66	0.0	0.0	0.0
16	Temparature -25 degree (T2)	-0.45	-0.65	0.0	0.0	0.0

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Combined Load Cases:		Load at top of foundation					Load at bottom of foundation				
Load Case	Load Combination	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
Allowable Stress Design											
101	D+Cl	18.6	2.2	0.0	0.0	0.0	139.71	2.18	4.36	0.00	0.00
102	D+Cl+Lr+Cr	22.7	2.5	0.0	0.0	0.0	143.83	2.46	4.92	0.00	0.00
103	D+Cl+Cr+T1	18.0	2.7	0.0	0.0	0.0	139.11	2.68	5.36	0.00	0.00
104	D+Cl+Cr+T2	17.1	1.4	0.0	0.0	0.0	138.21	1.37	2.74	0.00	0.00
105	D+Cl+Cr+0.75Lr+0.75T1	21.7	2.8	0.0	0.0	0.0	142.88	2.85	5.69	0.00	0.00
106	D+Cl+Cr+0.75Lr+0.75T2	21.1	1.9	0.0	0.0	0.0	142.20	1.86	3.73	0.00	0.00
107	D+Cl+WL1	-6.3	-13.1	0.0	0.0	0.0	114.84	-13.08	-26.16	0.00	0.00
108	D+Cl+WL2	-0.9	-17.0	0.0	0.0	0.0	120.25	-17.03	-34.06	0.00	0.00
109	D+Cl+WR1	15.1	14.2	0.0	0.0	0.0	136.20	14.17	28.34	0.00	0.00
110	D+Cl+WR2	20.6	10.3	0.0	0.0	0.0	141.68	10.26	20.52	0.00	0.00
111	D+Cl+WLn1	-29.6	8.1	0.0	-20.0	0.0	91.58	8.07	16.14	-20.00	-40.00
112	D+Cl+W_Ln2	-24.1	4.1	0.0	-20.0	0.0	97.05	4.14	8.28	-20.00	-40.00
113	D+Cl+W_Ln-1	-25.8	8.1	0.0	20.0	0.0	95.34	8.10	16.20	20.00	40.00
114	D+Cl+W_Ln-2	-20.3	4.2	0.0	20.0	0.0	100.80	4.18	8.36	20.00	40.00
115	D+Cl+0.7SL	13.0	1.9	0.0	7.0	0.0	134.08	1.90	3.80	7.00	14.00
116	D+Cl+0.7SR	24.2	2.5	0.0	-7.0	0.0	145.34	2.46	4.92	-7.00	-14.00
117	D+0.75Lr+0.75WL1	-6.6	-10.7	0.0	0.0	0.0	114.58	-10.72	-21.43	0.00	0.00
118	D+0.75Lr+0.75WL2	-2.5	-13.7	0.0	0.0	0.0	118.63	-13.68	-27.36	0.00	0.00
119	D+0.75Lr+0.75WR1	9.5	9.7	0.0	0.0	0.0	130.60	9.72	19.45	0.00	0.00
120	D+0.75Lr+0.75WR2	13.6	6.8	0.0	0.0	0.0	134.71	6.79	13.58	0.00	0.00
121	D+0.75Lr+0.75W_Ln1	-24.0	5.1	0.0	-15.0	0.0	97.13	5.15	10.30	-15.00	-30.00
122	D+0.75Lr+0.75W_Ln2	-19.9	2.2	0.0	-15.0	0.0	101.23	2.20	4.40	-15.00	-30.00
123	D+0.75Lr+0.75W_Ln-1	-21.2	5.2	0.0	15.0	0.0	99.95	5.17	10.34	15.00	30.00
124	D+0.75Lr+0.75W_Ln-2	-17.1	2.2	0.0	15.0	0.0	104.05	2.23	4.46	15.00	30.00
125	D+0.75Lr+0.525SL	7.9	0.5	0.0	5.3	0.0	129.01	0.52	1.04	5.25	10.50
126	D+0.75Lr+0.525SR	16.3	0.9	0.0	-5.3	0.0	137.45	0.94	1.88	-5.25	-10.50
127	0.6D+WL1	-19.9	-15.0	0.0	0.0	0.0	101.19	-15.02	-30.04	0.00	0.00
128	0.6D+WL2	-14.5	-19.0	0.0	0.0	0.0	106.60	-18.97	-37.94	0.00	0.00
129	0.6D+WR1	1.4	12.2	0.0	0.0	0.0	122.55	12.23	24.46	0.00	0.00
130	0.6D+WR2	6.9	8.3	0.0	0.0	0.0	128.03	8.32	16.64	0.00	0.00
131	0.6D+W_Ln1	-43.2	6.1	0.0	-20.0	0.0	77.93	6.13	12.26	-20.00	-40.00
132	0.6D+W_Ln2	-37.7	2.2	0.0	-20.0	0.0	83.40	2.20	4.40	-20.00	-40.00
133	0.6D+W_Ln-1	-39.4	6.2	0.0	20.0	0.0	81.69	6.16	12.32	20.00	40.00
134	0.6D+W_Ln-2	-34.0	2.2	0.0	20.0	0.0	87.15	2.24	4.48	20.00	40.00
135	0.6D+0.7SL	-0.696	-0.04	0	7	0	120.43	-0.04	-0.08	7.00	14.00
136	0.6D+0.7SR	10.56	0.52	0	-7	0	131.69	0.52	1.04	-7.00	-14.00

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Ultimate Conditions											
201	1.4(D+Cl)	26.0	3.1	0.0	0.0	0.0	147.14	3.05	6.10	0.00	0.00
202	1.4(D+Cl+Cr)	24.5	2.8	0.0	0.0	0.0	145.67	2.83	5.66	0.00	0.00
203	1.4(D+Cl+T1)+0.5Lr	29.2	4.2	0.0	0.0	0.0	150.36	4.20	8.39	0.00	0.00
204	1.4(D+Cl+T2)+0.5Lr	28.0	2.4	0.0	0.0	0.0	149.10	2.36	4.72	0.00	0.00
205	1.2(D+Cl)+1.6Lr+0.8WL1	10.7	-8.9	0.0	0.0	0.0	131.80	-8.89	-17.78	0.00	0.00
206	1.2(D+Cl)+1.6Lr+0.8WL2	15.0	-12.0	0.0	0.0	0.0	136.13	-12.05	-24.10	0.00	0.00
207	1.2(D+Cl)+1.6Lr+0.8WR1	27.8	12.9	0.0	0.0	0.0	148.89	12.91	25.82	0.00	0.00
208	1.2(D+Cl)+1.6Lr+0.8WR2	32.1	9.8	0.0	0.0	0.0	153.27	9.78	19.57	0.00	0.00
209	1.2(D+Cl)+1.6Lr+0.8W_Ln1	-7.9	8.0	0.0	-16.0	0.0	113.19	8.03	16.06	-16.00	-32.00
210	1.2(D+Cl)+1.6Lr+0.8W_Ln2	-3.6	4.9	0.0	-16.0	0.0	117.57	4.89	9.78	-16.00	-32.00
211	1.2(D+Cl)+1.6Lr+0.8W_Ln-1	-4.9	8.1	0.0	16.0	0.0	116.20	8.06	16.11	16.00	32.00
212	1.2(D+Cl)+1.6Lr+0.8W_Ln-2	-0.6	4.9	0.0	16.0	0.0	120.57	4.92	9.84	16.00	32.00
213	1.2(D+Cl)+0.5Lr+1.6WL1	-14.91	-21.58	0	0	0	106.22	-21.58	-43.16	0.00	0.00
214	1.2(D+Cl)+0.5Lr+1.6WR1	-6.26	-27.90	0	0	0	114.88	-27.90	-55.80	0.00	0.00
215	1.2(D+Cl)+0.5Lr+1.6WL2	19.27	22.02	0	0	0	140.40	22.02	44.04	0.00	0.00
216	1.2(D+Cl)+0.5Lr+1.6WR2	28.03	15.76	0	0	0	149.16	15.76	31.53	0.00	0.00
217	1.2(D+Cl)+0.5Lr+1.6W_Ln1	-52.13	12.26	0	-32	0	69.00	12.26	24.52	-32.00	-64.00
218	1.2(D+Cl)+0.5Lr+1.6W_Ln2	-43.38	5.97	0	-32	0	77.76	5.97	11.94	-32.00	-64.00
219	1.2(D+Cl)+0.5Lr+1.6W_Ln-1	-46.11	12.31	0	32	0	75.02	12.31	24.62	32.00	64.00
220	1.2(D+Cl)+0.5Lr+1.6W_Ln-2	-37.38	6.04	0	32	0	83.76	6.04	12.07	32.00	64.00
221	1.2D+SL	1.82	0.08	0	10	0	122.95	0.08	0.16	10.00	20.00
222	1.2D+SR	17.90	0.88	0	-10	0	139.03	0.88	1.76	-10.00	-20.00
223	0.9D+1.6WL1	-32.39	-24.06	0	0	0	88.74	-24.06	-48.11	0.00	0.00
224	0.9D+1.6WL2	-23.74	-30.38	0	0	0	97.39	-30.38	-60.75	0.00	0.00
225	0.9D+1.6WR1	1.78	19.54	0	0	0	122.91	19.54	39.09	0.00	0.00
226	0.9D+1.6WR2	10.55	13.29	0	0	0	131.68	13.29	26.58	0.00	0.00
227	0.9D+1.6W_Ln1	-69.61	9.78	0	-32	0	51.52	9.78	19.57	-32.00	-64.00
228	0.9D+1.6W_Ln2	-60.86	3.50	0	-32	0	60.27	3.50	6.99	-32.00	-64.00
229	0.9D+1.6W_Ln-1	-63.59	9.83	0	32	0	57.54	9.83	19.66	32.00	64.00
230	0.9D+1.6W_Ln-2	-54.86	3.56	0	32	0	66.27	3.56	7.12	32.00	64.00
231	0.9D+1.0SL	-0.64	-0.04	0	10	0	120.49	-0.04	-0.08	10.00	20.00
232	0.9D+1.0SR	15.44	0.76	0	-10	0	136.57	0.76	1.52	-10.00	-20.00

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Check for overturning and sliding:

	Load Cases:	Load at b/o fdn					F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
		V (kN)	Hx (kN)	Mz (kNm)	Hz (kN)	Mx (kNm)				
101	D+Cl	139.71	2.18	4.36	0.00	0.00	32.04	NA	25.63	NA
102	D+Cl+Lr+Cr	143.83	2.46	4.92	0.00	1.00	29.23	144	23.39	NA
103	D+Cl+Cr+T1	139.11	2.68	5.36	0.00	2.00	25.95	70	20.76	NA
104	D+Cl+Cr+T2	138.21	1.37	2.74	0.00	3.00	50.44	46	40.35	NA
105	D+Cl+Cr+0.75Lr+0.75T1	142.88	2.85	5.69	0.00	4.00	25.11	36	20.09	NA
106	D+Cl+Cr+0.75Lr+0.75T2	142.20	1.86	3.73	0.00	5.00	38.17	28	30.54	NA
107	D+Cl+WL1	114.84	-13.08	-26.16	0.00	6.00	4.39	19	3.51	NA
108	D+Cl+WL2	120.25	-17.03	-34.06	0.00	7.00	3.53	17	2.82	NA
109	D+Cl+WR1	136.20	14.17	28.34	0.00	8.00	4.81	17	3.84	NA
110	D+Cl+WR2	141.68	10.26	20.52	0.00	9.00	6.90	16	5.52	NA
111	D+Cl+WLn1	91.58	8.07	16.14	-20.00	10.00	5.67	9	4.54	1.83
112	D+Cl+W_Ln2	97.05	4.14	8.28	-20.00	11.00	11.72	9	9.38	1.94
113	D+Cl+W_Ln-1	95.34	8.10	16.20	20.00	12.00	5.89	8	4.71	1.91
114	D+Cl+W_Ln-2	100.80	4.18	8.36	20.00	13.00	12.06	8	9.65	2.02
115	D+Cl+0.7SL	134.08	1.90	3.80	7.00	14.00	35.28	10	28.23	7.66
116	D+Cl+0.7SR	145.34	2.46	4.92	-7.00	15.00	29.54	10	23.63	8.31
117	D+0.75Lr+0.75WL1	114.58	-10.72	-21.43	0.00	16.00	5.35	7	4.28	NA
118	D+0.75Lr+0.75WL2	118.63	-13.68	-27.36	0.00	17.00	4.34	7	3.47	NA
119	D+0.75Lr+0.75WR1	130.60	9.72	19.45	0.00	18.00	6.72	7	5.37	NA
120	D+0.75Lr+0.75WR2	134.71	6.79	13.58	0.00	19.00	9.92	7	7.94	NA
121	D+0.75Lr+0.75W_Ln1	97.13	5.15	10.30	-15.00	20.00	9.43	5	7.55	2.59
122	D+0.75Lr+0.75W_Ln2	101.23	2.20	4.40	-15.00	21.00	23.01	5	18.41	2.70
123	D+0.75Lr+0.75W_Ln-1	99.95	5.17	10.34	15.00	22.00	9.67	5	7.73	2.67
124	D+0.75Lr+0.75W_Ln-2	104.05	2.23	4.46	15.00	23.00	23.33	5	18.66	2.77
125	D+0.75Lr+0.525SL	129.01	0.52	1.04	5.25	24.00	124.04	5	99.24	9.83
126	D+0.75Lr+0.525SR	137.45	0.94	1.88	-5.25	25.00	73.11	5	58.49	10.47
127	0.6D+WL1	101.19	-15.02	-30.04	0.00	26.00	3.37	4	2.69	NA
128	0.6D+WL2	106.60	-18.97	-37.94	0.00	27.00	2.81	4	2.25	NA
129	0.6D+WR1	122.55	12.23	24.46	0.00	28.00	5.01	4	4.01	NA
130	0.6D+WR2	128.03	8.32	16.64	0.00	29.00	7.69	4	6.16	NA
131	0.6D+W_Ln1	77.93	6.13	12.26	-20.00	30.00	6.36	3	5.09	1.56
132	0.6D+W_Ln2	83.40	2.20	4.40	-20.00	31.00	18.96	3	15.16	1.67
133	0.6D+W_Ln-1	81.69	6.16	12.32	20.00	32.00	6.63	3	5.30	1.63
134	0.6D+W_Ln-2	87.15	2.24	4.48	20.00	33.00	19.45	3	15.56	1.74
135	0.6D+0.7SL	120.43	-0.04	-0.08	7.00	34.00	1505.43	4	1204.34	6.88
136	0.6D+0.7SR	131.69	0.52	1.04	-7.00	35.00	126.63	4	101.30	7.53

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FOUNDATION MKD. F1

Load Combination & Calculation of Base pressure

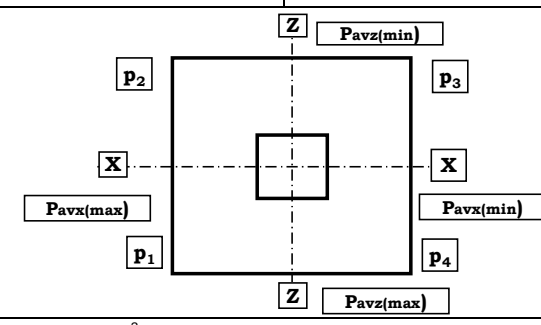
Max. Base pressure = $p_{max} = (V/(Lx.Lz)) + (6.Mz/((Lx)^2.Lz) + (6.Mx/(Lx.(Lz)^2)$

Min. Base pressure = $p_{min} = (V/(Lx.Lz)) - (6.Mz/((Lx)^2.Lz) - (6.Mx/(Lx.(Lz)^2)$

Surcharge load = 20 kN/m^2

Net SBC of soil for DL, LL= 400 kN/m^2 For WL/SL case = 500 kN/m^2

Gross SBC of soil for DL, LL= 427 kN/m^2 For WL/SL case = 527 kN/m^2



Load Case	Load Combination	Design Load/moments			Base Pressure (kN/m^2)												
		V (kN)	Mz (kN-m)	Mx (kN-m)	V/ (Lx.Lz)	6.Mz/ $(Lx)^2$.Lz	6.Mx/ Lx.$(Lz)^2$	p_1 (p_{max})	p_2	p_4	p_3 (p_{min})	(p_{max}-γh)	$p_{avx(max)}$	$p_{avx(min)}$	$p_{avz(max)}$	$p_{avz(min)}$	Remarks
101	D+Cl	139.71	4.36	0.00	34.93	3.27	0.00	38.20	38.20	31.66	31.66	11.20	38.20	31.66	34.93	34.93	O.K
102	D+Cl+Lr+Cr	143.83	4.92	0.00	35.96	3.69	0.00	39.65	39.65	32.27	32.27	12.65	39.65	32.27	35.96	35.96	O.K
103	D+Cl+Cr+T1	139.11	5.36	0.00	34.78	4.02	0.00	38.80	38.80	30.76	30.76	11.80	38.80	30.76	34.78	34.78	O.K
104	D+Cl+Cr+T2	138.21	2.74	0.00	34.55	2.06	0.00	36.61	36.61	32.50	32.50	9.61	36.61	32.50	34.56	34.56	O.K
105	D+Cl+Cr+0.75Lr+0.75T1	142.88	5.69	0.00	35.72	4.27	0.00	39.99	39.99	31.45	31.45	12.99	39.99	31.45	35.72	35.72	O.K
106	D+Cl+Cr+0.75Lr+0.75T2	142.20	3.73	0.00	35.55	2.79	0.00	38.34	38.34	32.76	32.76	11.34	38.34	32.76	35.55	35.55	O.K
107	D+Cl+WL1	114.84	26.16	0.00	28.71	19.62	0.00	48.33	48.33	9.09	9.09	21.33	48.33	9.09	28.71	28.71	O.K
108	D+Cl+WL2	120.25	34.06	0.00	30.06	25.55	0.00	55.61	55.61	4.52	4.52	28.61	55.61	4.52	30.07	30.07	O.K
109	D+Cl+WR1	136.20	28.34	0.00	34.05	21.26	0.00	55.31	55.31	12.80	12.80	28.31	55.31	12.80	34.06	34.06	O.K
110	D+Cl+WR2	141.68	20.52	0.00	35.42	15.39	0.00	50.81	50.81	20.03	20.03	23.81	50.81	20.03	35.42	35.42	O.K
111	D+Cl+WLn1	91.58	16.14	40.00	22.90	12.11	30.00	65.00	5.00	40.79	-19.21	38.00	35.00	10.79	52.90	-7.11	O.K
112	D+Cl+W_Ln2	97.05	8.28	40.00	24.26	6.21	30.00	60.47	0.47	48.05	-11.95	33.47	30.47	18.05	54.26	-5.74	O.K
113	D+Cl+W_Ln-1	95.34	16.20	40.00	23.84	12.15	30.00	65.99	5.99	41.69	-18.32	38.99	35.99	11.69	53.84	-6.17	O.K
114	D+Cl+W_Ln-2	100.80	8.36	40.00	25.20	6.27	30.00	61.47	1.47	48.93	-11.07	34.47	31.47	18.93	55.20	-4.80	O.K
115	D+Cl+0.7SL	134.08	3.80	14.00	33.52	2.85	10.50	46.87	25.87	41.17	20.17	19.87	36.37	30.67	44.02	23.02	O.K
116	D+Cl+0.7SR	145.34	4.92	14.00	36.33	3.69	10.50	50.52	29.52	43.14	22.14	23.52	40.02	32.64	46.83	25.83	O.K
117	D+0.75Lr+0.75WL1	114.58	21.43	0.00	28.64	16.07	0.00	44.72	44.72	12.57	12.57	17.72	44.72	12.57	28.65	28.65	O.K
118	D+0.75Lr+0.75WL2	118.63	27.36	0.00	29.66	20.52	0.00	50.17	50.17	9.14	9.14	23.17	50.17	9.14	29.66	29.66	O.K
119	D+0.75Lr+0.75WR1	130.60	19.45	0.00	32.65	14.58	0.00	47.23	47.23	18.07	18.07	20.23	47.23	18.07	32.65	32.65	O.K
120	D+0.75Lr+0.75WR2	134.71	13.58	0.00	33.68	10.19	0.00	43.86	43.86	23.49	23.49	16.86	43.86	23.49	33.68	33.68	O.K
121	D+0.75Lr+0.75W_Ln1	97.13	10.30	30.00	24.28	7.72	22.50	54.50	9.50	39.06	-5.94	27.50	32.00	16.56	46.78	1.78	O.K
122	D+0.75Lr+0.75W_Ln2	101.23	4.40	30.00	25.31	3.30	22.50	51.11	6.11	44.51	-0.49	24.11	28.61	22.01	47.81	2.81	O.K
123	D+0.75Lr+0.75W_Ln-1	99.95	10.34	30.00	24.99	7.76	22.50	55.24	10.24	39.73	-5.27	28.24	32.74	17.23	47.49	2.49	O.K
124	D+0.75Lr+0.75W_Ln-2	104.05	4.46	30.00	26.01	3.35	22.50	51.86	6.86	45.17	0.17	24.86	29.36	22.67	48.52	3.52	O.K
125	D+0.75Lr+0.525SL	129.01	1.04	10.50	32.25	0.78	7.88	40.91	25.16	39.35	23.60	13.91	33.04	31.48	40.13	24.38	O.K
126	D+0.75Lr+0.525SR	137.45	1.88	10.50	34.36	1.41	7.88	43.65	27.90	40.83	25.08	16.65	35.78	32.96	42.24	26.49	O.K
127	0.6D+WL1	101.19	30.04	0.00	25.30	22.53	0.00	47.83	47.83	2.77	2.77	20.83	47.83	2.77	25.30	25.30	O.K
128	0.6D+WL2	106.60	37.94	0.00	26.65	28.46	0.00	55.11	55.11	-1.80	-1.80	28.11	55.11	-1.80	26.66	26.66	O.K
129	0.6D+WR1	122.55	24.46	0.00	30.64	18.35	0.00	48.98	48.98	12.29	12.29	21.98	48.98	12.29	30.64	30.64	O.K
130	0.6D+WR2	128.03	16.64	0.00	32.01	12.48	0.00	44.49	44.49	19.53	19.53	17.49	44.49	19.53	32.01	32.01	O.K
131	0.6D+W_Ln1	77.93	12.26	40.00	19.48	9.20	30.00	58.68	-1.32	40.29	-19.71	31.68	28.68	10.29	49.49	-10.52	O.K
132	0.6D+W_Ln2	83.40	4.40	40.00	20.85	3.30	30.00	54.15	-5.85	47.55	-12.45	27.15	24.15	17.55	50.85	-9.15	O.K
133	0.6D+W_Ln-1	81.69	12.32	40.00	20.42	9.24	30.00	59.66	-0.34	41.18	-18.82	32.66	29.66	11.18	50.42	-9.58	O.K
134	0.6D+W_Ln-2	87.15	4.48	40.00	21.79	3.36	30.00	55.15	-4.85	48.43	-11.57	28.15	25.15	18.43	51.79	-8.21	O.K
135	0.6D+0.7SL	120.43	0.08	14.00	30.11	0.06	10.50	40.67	19.67	40.55	19.55	13.67	30.17	30.05	40.61	19.61	O.K
136	0.6D+0.7SR	131.69	1.04	14.00	32.92	0.78	10.50	44.20	23.20	42.64	21.64	17.20	33.70	32.14	43.42	22.42	O.K

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201	1.4(D+Cl)	147.14	6.10	0.00	36.79	4.58	0.00	41.36	41.36	32.21	32.21	14.36	41.36	32.21	36.79	36.79
202	1.4(D+Cl+Cr)	145.67	5.66	0.00	36.42	4.24	0.00	40.66	40.66	32.18	32.18	13.66	40.66	32.18	36.42	36.42
203	1.4(D+Cl+T1)+0.5Lr	150.36	8.39	0.00	37.59	6.29	0.00	43.88	43.88	31.30	31.30	16.88	43.88	31.30	37.59	37.59
204	1.4(D+Cl+T2)+0.5Lr	149.10	4.72	0.00	37.27	3.54	0.00	40.82	40.82	33.73	33.73	13.82	40.82	33.73	37.28	37.28
205	1.2(D+Cl)+1.6Lr+0.8WL1	131.80	17.78	0.00	32.95	13.33	0.00	46.28	46.28	19.62	19.62	19.28	46.28	19.62	32.95	32.95
206	1.2(D+Cl)+1.6Lr+0.8WL2	136.13	24.10	0.00	34.03	18.07	0.00	52.10	52.10	15.96	15.96	25.10	52.10	15.96	34.03	34.03
207	1.2(D+Cl)+1.6Lr+0.8WR1	148.89	25.82	0.00	37.22	19.37	0.00	56.59	56.59	17.85	17.85	29.59	56.59	17.85	37.22	37.22
208	1.2(D+Cl)+1.6Lr+0.8WR2	153.27	19.57	0.00	38.32	14.68	0.00	52.99	52.99	23.64	23.64	25.99	52.99	23.64	38.32	38.32
209	1.2(D+Cl)+1.6Lr+0.8W_L	113.19	16.06	32.00	28.30	12.05	24.00	64.35	16.35	40.25	-7.75	37.35	40.35	16.25	52.30	4.30
210	1.2(D+Cl)+1.6Lr+0.8W_L	117.57	9.78	32.00	29.39	7.33	24.00	60.72	12.72	46.06	-1.94	33.72	36.72	22.06	53.39	5.39
211	1.2(D+Cl)+1.6Lr+0.8W_L	116.20	16.11	32.00	29.05	12.08	24.00	65.13	17.13	40.97	-7.03	38.13	41.13	16.97	53.05	5.05
212	1.2(D+Cl)+1.6Lr+0.8W_L	120.57	9.84	32.00	30.14	7.38	24.00	61.52	13.52	46.76	-1.24	34.52	37.52	22.76	54.14	6.14
213	1.2(D+Cl)+0.5Lr+1.6WL1	106.22	43.16	0.00	26.55	32.37	0.00	58.92	58.92	-5.82	-5.82	31.92	58.92	-5.82	26.55	26.55
214	1.2(D+Cl)+0.5Lr+1.6WR1	114.88	55.80	0.00	28.72	41.85	0.00	70.57	70.57	-13.13	-13.13	43.57	70.57	-13.13	28.72	28.72
215	1.2(D+Cl)+0.5Lr+1.6WL2	140.40	44.04	0.00	35.10	33.03	0.00	68.13	68.13	2.07	2.07	41.13	68.13	2.07	35.10	35.10
216	1.2(D+Cl)+0.5Lr+1.6WR2	149.16	31.53	0.00	37.29	23.65	0.00	60.94	60.94	13.64	13.64	33.94	60.94	13.64	37.29	37.29
217	1.2(D+Cl)+0.5Lr+1.6W_L	69.00	24.52	64.00	17.25	18.39	48.00	83.64	-12.36	46.86	-49.14	56.64	35.64	-1.14	65.25	-30.75
218	1.2(D+Cl)+0.5Lr+1.6W_L	77.76	11.94	64.00	19.44	8.96	48.00	76.40	-19.60	58.48	-37.52	49.40	28.40	10.48	67.44	-28.56
219	1.2(D+Cl)+0.5Lr+1.6W_L	75.02	24.62	64.00	18.75	18.46	48.00	85.22	-10.78	48.29	-47.71	58.22	37.22	0.29	66.76	-29.25
220	1.2(D+Cl)+0.5Lr+1.6W_L	83.76	12.07	64.00	20.94	9.05	48.00	77.99	-18.01	59.88	-36.12	50.99	29.99	11.88	68.94	-27.07
221	1.2D+SL	122.95	0.16	20.00	30.74	0.12	15.00	45.86	15.86	45.62	15.62	18.86	30.86	30.62	45.74	15.74
222	1.2D+SR	139.03	1.76	20.00	34.76	1.32	15.00	51.08	21.08	48.44	18.44	24.08	36.08	33.44	49.76	19.76
223	0.9D+1.6WL1	88.74	48.11	0.00	22.18	36.08	0.00	58.27	58.27	-13.90	-13.90	31.27	58.27	-13.90	22.19	22.19
224	0.9D+1.6WL2	97.39	60.75	0.00	24.35	45.56	0.00	69.91	69.91	-21.22	-21.22	42.91	69.91	-21.22	24.35	24.35
225	0.9D+1.6WR1	122.91	39.09	0.00	30.73	29.32	0.00	60.04	60.04	1.41	1.41	33.04	60.04	1.41	30.73	30.73
226	0.9D+1.6WR2	131.68	26.58	0.00	32.92	19.93	0.00	52.85	52.85	12.99	12.99	25.85	52.85	12.99	32.92	32.92
227	0.9D+1.6W_Ln1	51.52	19.57	64.00	12.88	14.68	48.00	75.56	-20.44	46.20	-49.80	48.56	27.56	-1.80	60.88	-35.12
228	0.9D+1.6W_Ln2	60.27	6.99	64.00	15.07	5.24	48.00	68.31	-27.69	57.82	-38.18	41.31	20.31	9.82	63.07	-32.94
229	0.9D+1.6W_Ln-1	57.54	19.66	64.00	14.38	14.75	48.00	77.13	-18.87	47.64	-48.36	50.13	29.13	-0.36	62.39	-33.62
230	0.9D+1.6W_Ln-2	66.27	7.12	64.00	16.57	5.34	48.00	69.91	-26.09	59.23	-36.77	42.91	21.91	11.23	64.57	-31.43
231	0.9D+1.0SL	120.49	0.08	20.00	30.12	0.06	15.00	45.18	15.18	45.06	15.06	18.18	30.18	30.06	45.12	15.12
232	0.9D+1.0SR	136.57	1.52	20.00	34.14	1.14	15.00	50.28	20.28	48.00	18.00	23.28	35.28	33.00	49.14	19.14
	Max :	153	61	64												
	Min :	52	0	0												

Max pressure at base of foundation without surcharge excluding weight of soil (DL+LL cases) = 12.99 kN/m^2

Max pressure at base of foundation including surcharge excluding weight of soil (DL+LL cases) = 32.99 kN/m^2

Max pressure at base of foundation without surcharge excluding weight of soil (DL+LL+SL/WL cases) = 58.22 kN/m^2

Max pressure at base of foundation including surcharge excluding weight of soil (DL+LL+WL/SL cases) = 78.22 kN/m^2

Sample calculation of values of base pressure:

Load case : D+Cl+Lr+Cr

Maximum pressure, p_1 (pmax) = 35.96+3.69+0.00= 39.65 kN/m^2

p_2= 35.96+3.69-0.00= 39.65 kN/m^2

p_3= 35.96-(3.69+0.00)= 32.27 kN/m^2

p_4= 35.96-3.69+0.00= 32.27 kN/m^2

Maxm pressure excluding weight of soil (p_{max}-γh)= 39.65-1.5*18= 12.65 kN/m^2

Maximum average Pressure along X, $p_{avx(max)}$ = (39.65+39.65)/2= 39.65 kN/m^2

Minimum average Pressure along X, $p_{avx(minx)}$ = (32.27+32.27)/2= 32.27 kN/m^2

Maximum average Pressure along Y, $p_{avy(max)}$ = (39.65+32.27)/2= 35.96 kN/m^2

Minimum average Pressure along Y, $p_{avy(minx)}$ = (32.27+39.65)/2= 35.96 kN/m^2

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FOUNDATION MKD. F1

Calculation of Bending Moment and Shear Force in X direction :

p_{1x} =pressure at the face of column at section 1x-1x in kN/m^2.

p_{2x} =pressure at the face of column at section 2x-2x in kN/m^2.

p_{3x} =pressure at a distance 'd' from the face of column at section 3x-3x in kN/m^2.

BM_{1x} =Moment at the face of column at section 1x-1x in kN-m/m.

BM_{2x} =Moment at the face of column 2x-2x in kN-m/m.

SF_{3x} = Oneway shear force at a distance 'd' from the face of column at section 3x-3x in kN/m.

SF (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Punching perimeter= ((0.5+2*0.5*0.282)+(0.5+2*0.5*0.282))*2

= 3.128 m

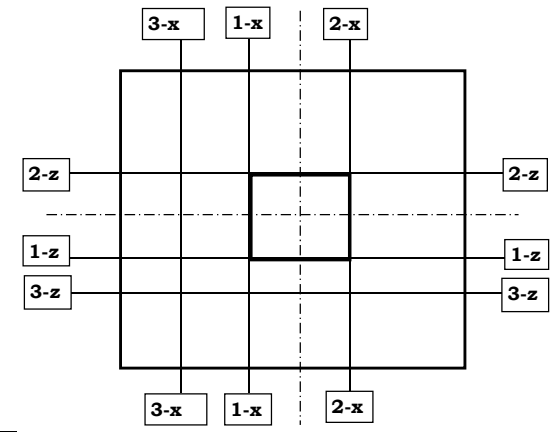
Punching Area= ((0.5+2*0.5*0.282)*(0.5+2*0.5*0.282))

= 0.61 m^2 Punching shear stress to be checked

Downward pressure due to self weight of raft and soil = (0.35*24+1.15*18)

= 29.10 kN/m^2

Area of footing, A_f= 4.00 m^2



Load Case	Load Combination	p_{1x} kN/m2	p_{2x} kN/m2	p_{3x} kN/m2	$B.M_{1x}$ kNm	$B.M_{2x}$ kNm	$S.F_{3x}$ kN	$S.F_{(punching)}$ kN
Ultimate Condition								
201	1.4(D+Cl)	37.93	35.64	39.27	3.13	1.20	5.11	26.05
202	1.4(D+Cl+Cr)	37.48	35.36	38.73	2.95	1.16	4.83	24.81
203	1.4(D+Cl+T1)+0.5Lr	39.16	36.02	41.01	3.71	1.06	6.09	28.78
204	1.4(D+Cl+T2)+0.5Lr	38.16	36.39	39.20	3.05	1.55	4.98	27.71
205	1.2(D+Cl)+1.6Lr+0.8WL1	36.28	29.62	40.20	3.89	-1.73	6.45	13.05
206	1.2(D+Cl)+1.6Lr+0.8WL2	38.55	29.51	43.86	5.20	-2.43	8.61	16.72
207	1.2(D+Cl)+1.6Lr+0.8WR1	42.06	32.38	47.76	6.37	-1.80	10.52	27.54
208	1.2(D+Cl)+1.6Lr+0.8WR2	41.98	34.65	46.30	5.69	-0.50	9.37	31.25
209	1.2(D+Cl)+1.6Lr+0.8W_Ln1	31.31	25.29	34.86	2.32	-2.77	3.88	-2.72
210	1.2(D+Cl)+1.6Lr+0.8W_Ln2	31.22	27.56	33.38	1.63	-1.46	2.71	0.99
211	1.2(D+Cl)+1.6Lr+0.8W_Ln-	32.07	26.03	35.62	2.53	-2.56	4.23	-0.17
212	1.2(D+Cl)+1.6Lr+0.8W_Ln-	31.99	28.30	34.15	1.85	-1.26	3.07	3.53
213	1.2(D+Cl)+0.5Lr+1.6WL1	34.64	18.46	44.16	6.11	-7.55	10.23	-8.63
214	1.2(D+Cl)+0.5Lr+1.6WR1	39.18	18.26	51.49	8.72	-8.93	14.56	-1.29
215	1.2(D+Cl)+0.5Lr+1.6WL2	43.36	26.84	53.07	8.65	-5.28	14.36	20.34
216	1.2(D+Cl)+0.5Lr+1.6WR2	43.20	31.38	50.16	7.29	-2.69	12.06	27.77
217	1.2(D+Cl)+0.5Lr+1.6W_Ln1	21.85	12.65	27.25	0.55	-7.21	1.07	-40.17
218	1.2(D+Cl)+0.5Lr+1.6W_Ln2	21.68	17.20	24.31	-0.83	-4.61	-1.25	-32.75
219	1.2(D+Cl)+0.5Lr+1.6W_Ln-	23.37	14.14	28.80	0.99	-6.80	1.78	-35.07
220	1.2(D+Cl)+0.5Lr+1.6W_Ln-	23.20	18.67	25.86	-0.39	-4.21	-0.54	-27.67
221	1.2D+SL	30.77	30.71	30.81	0.49	0.44	0.79	5.55
222	1.2D+SR	35.09	34.43	35.48	1.87	1.31	3.05	19.18
223	0.9D+1.6WL1	31.21	13.16	41.82	5.67	-9.56	9.55	-23.45
224	0.9D+1.6WL2	35.74	12.95	49.13	8.27	-10.95	13.87	-16.11
225	0.9D+1.6WR1	38.05	23.40	46.67	6.64	-5.73	11.06	5.52
226	0.9D+1.6WR2	37.90	27.94	43.76	5.28	-3.13	8.76	12.95
227	0.9D+1.6W_Ln1	16.55	9.21	20.87	-1.47	-7.66	-2.23	-54.99
228	0.9D+1.6W_Ln2	16.38	13.75	17.92	-2.84	-5.05	-4.55	-47.57
229	0.9D+1.6W_Ln-1	18.07	10.70	22.41	-1.03	-7.25	-1.52	-49.89
230	0.9D+1.6W_Ln-2	17.91	15.24	19.47	-2.40	-4.65	-3.83	-42.48
231	0.9D+1.0SL	30.14	30.11	30.15	0.30	0.27	0.49	3.46
232	0.9D+1.0SR	34.43	33.86	34.76	1.66	1.18	2.70	17.09

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Sample calculation of BM & SF

Load case : 201

p_{1x} = 41.36-(41.36-32.21)*0.75/2 = 37.93 kN/m^2 Overhang = 0.75 m

p_{2x} = 32.21+(41.36-32.21)*0.75/2 = 35.64 kN/m^2 Effective depth = 0.294 m

p_{3x} = 41.36-(41.36-32.21)*(0.75-0.294)/2 = 39.27 kN/m^2

BM_{1x} = (37.93-29.1)*0.75*0.75/2+0.5*(41.36-37.93)*0.75*0.75*0.67 = 3.13 kN-m/m

BM_{2x} = (35.64-29.1)*0.75*0.75/2-0.5*(-32.21+35.64)*0.75*0.75*0.67 = 1.20 kN-m/m

SF_{3x} = (39.27-29.1)*(0.75-0.294)+0.5*(41.36-39.27)*(0.75-0.294) = 5.11 kN/m

$S.F_{(punching)}$ = (4-0.61)*(36.79-29.1) = 26.05 kN

Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	8.72	0.10	0.13	0.000	94.38	529.20	12ϕT	213.71	300	200	565.49
At top	-10.95	-0.13	0.17	0.000	118.56	529.20	12ϕT	213.71	300	200	565.49

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, $(\phi\ Vc)/bd$ = 0.740 N/mm2

Shear at a distance 'd' from face of column, Vu= 6.45 kN/m

Shear stress , tv= (6.45*1000)/(1000*294)

= 0.02 N/mm2 **O.K.**

Check against Two-way shear force at a distance d/2 from face of column:

Design punching shear at a distance d/2 from face of column, Vu= 28.78 kN

Strength of Concrete in shear Vc1 = 2720.31 kN Table 22.6.5.2_ (a)

Strength of Concrete in shear Vc2 = 1759.04 kN Table 22.6.5.2_ (b)

Strength of Concrete in shear Vc3 = 1813.54 kN Table 22.6.5.2_ (c)

Design Vc= 1759.04 kN

Allowable shear stress, $(\phi\ Vc)/bd$ = 1.43 N/mm2

Design punching shear stress, τ = (28.78*1000)/(3128*294)

= 0.03 N/mm^2 **No shear reinforcement reqd.**

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Calculation of Bending Moment and Shear Force in Z direction :

p_{1z} =pressure at the face of column at section 1z-1z in kN/m^2.
p_{2z} =pressure at the face of column at section 2z-2z in kN/m^2.
p_{3z} =pressure at a distance d from the face of column at section 3z-3z in kN/m^2.
$B.M_{1z}$ =Moment at the face of column at section 1z-1z in kN-m/m.
$B.M_{2z}$ =Moment at the face of column at section 2z-2z in kN-m/m.
$S.F_{3z}$ = Oneway Shear Force at a distance d from the face of column at section 3z-3z in kN/m.
S.F (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Load Case	Load Combination	p_{1z} kN/m2	p_{2z} kN/m2	p_{3z} kN/m2	$B.M_{1z}$ kNm	$B.M_{2z}$ kNm	$S.F_{3z}$ kN
Ultimate Conditions							
201	1.4(D+Cl)	36.79	36.79	36.79	2.16	2.16	3.60
202	1.4(D+Cl+Cr)	36.42	36.42	36.42	2.06	2.06	3.43
203	1.4(D+Cl+T1)+0.5Lr	37.59	37.59	37.59	2.39	2.39	3.97
204	1.4(D+Cl+T2)+0.5Lr	37.28	37.28	37.28	2.30	2.30	3.83
205	1.2(D+Cl)+1.6Lr+0.8WL1	32.95	32.95	32.95	1.08	1.08	1.80
206	1.2(D+Cl)+1.6Lr+0.8WL2	34.03	34.03	34.03	1.39	1.39	2.31
207	1.2(D+Cl)+1.6Lr+0.8WR	37.22	37.22	37.22	2.28	2.28	3.80
208	1.2(D+Cl)+1.6Lr+0.8WR	38.32	38.32	38.32	2.59	2.59	4.31
209	1.2(D+Cl)+1.6Lr+0.8W_l	34.30	22.30	41.07	4.85	-5.30	8.23
210	1.2(D+Cl)+1.6Lr+0.8W_l	35.39	23.39	42.16	5.16	-5.00	8.74
211	1.2(D+Cl)+1.6Lr+0.8W_l	35.05	23.05	41.82	5.07	-5.09	8.58
212	1.2(D+Cl)+1.6Lr+0.8W_l	36.14	24.14	42.91	5.37	-4.79	9.09
213	1.2(D+Cl)+0.5Lr+1.6WL1	26.55	26.55	26.55	-0.72	-0.72	-1.19
214	1.2(D+Cl)+0.5Lr+1.6WR	28.72	28.72	28.72	-0.11	-0.11	-0.18
215	1.2(D+Cl)+0.5Lr+1.6WL2	35.10	35.10	35.10	1.69	1.69	2.81
216	1.2(D+Cl)+0.5Lr+1.6WR	37.29	37.29	37.29	2.30	2.30	3.83
217	1.2(D+Cl)+0.5Lr+1.6W_l	29.25	5.25	42.79	6.83	-13.49	11.66
218	1.2(D+Cl)+0.5Lr+1.6W_l	31.44	7.44	44.98	7.44	-12.88	12.69
219	1.2(D+Cl)+0.5Lr+1.6W_l	30.76	6.75	44.29	7.25	-13.07	12.37
220	1.2(D+Cl)+0.5Lr+1.6W_l	32.94	8.93	46.47	7.86	-12.46	13.39
221	1.2D+SL	34.49	26.99	38.72	3.64	-2.71	6.14
222	1.2D+SR	38.51	31.01	42.74	4.77	-1.58	8.03
223	0.9D+1.6WL1	22.19	22.19	22.19	-1.94	-1.94	-3.23
224	0.9D+1.6WL2	24.35	24.35	24.35	-1.34	-1.34	-2.22
225	0.9D+1.6WR1	30.73	30.73	30.73	0.46	0.46	0.76
226	0.9D+1.6WR2	32.92	32.92	32.92	1.07	1.07	1.79
227	0.9D+1.6W_Ln1	24.88	0.88	38.42	5.60	-14.72	9.62
228	0.9D+1.6W_Ln2	27.07	3.06	40.60	6.21	-14.11	10.64
229	0.9D+1.6W_Ln-1	26.39	2.38	39.92	6.02	-14.30	10.32
230	0.9D+1.6W_Ln-2	28.57	4.57	42.11	6.63	-13.68	11.34
231	0.9D+1.0SL	33.87	26.37	38.10	3.46	-2.89	5.85
232	0.9D+1.0SR	37.89	30.39	42.12	4.59	-1.76	7.74

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Sample calculation of BM & SF

Load case : 201

p_{1z} = 36.79-(36.79-36.79)*0.75/2= 36.79 kN/m^2

p_{2z} = 36.79+(36.79-36.79)*0.75/2= 36.79 kN/m^2

p_{3z} = 36.79-(36.79-36.79)*(0.75-0.282)/2= 36.79 kN/m^2

Overhang = 0.75 m

Effective depth = 0.282 m

BM_{1z} = (36.79-29.1)*0.75*0.75/2+0.5*(36.79-36.79)*0.75*0.75*0.67 = 2.16 kN-m/m

BM_{2z} = (36.79-29.1)*0.75*0.75/2-0.5*(-36.79+36.79)*0.75*0.75*0.67 = 2.16 kN-m/m

SF_{3z} = (36.79-29.1)*(0.75-0.282)+0.5*(36.79-36.79)*(0.75-0.282) = 3.60 kN/m

Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	7.86	0.099	0.132	0.0003	92.49	507.60	12ϕT	222.81	300	200	565.5
At top	-14.72	-0.185	-0.247	-0.0006	-1.65	507.60	12ϕT	222.81	300	200	565.5

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc) = 0.740 N/mm2

Shear at a distance 'd' from face of column, V_u= 3.97 kN/m

Shear stress , τ_v= (3.97*1000)/(1000*282)

= 0.01 N/mm2 **O.K.**

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :				
	At Bottom	12ϕT	200	c/c
	At Top	12ϕT	200	c/c
Along Z Direction :				
	At Bottom	12ϕT	200	c/c
	At Top	12ϕT	200	c/c

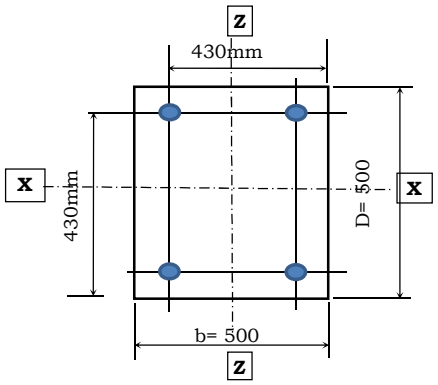
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5.2 Design calculation of column C1

COLUMN MKD C1



Input:

Clear cover to reinf.=	50mm	Table 20.6.1.3.1
Dia. of main reinf.=	20mm	
Dia. of link=	10mm	
Effective cover =	70mm	
Grade of Concrete (f_c') =	M35	
Grade of Steel (f_y) =	Fy420	
Grade of Reinf(links).=	Fy420	
Min. Steel Ratio	0.50%	
Max. Steel Ratio	4.00%	
Self weight of Column =	9.6kN	

For Compression-controlled sections

Strength reduction factor ϕ = 0.65 Table 21.2.2

Unsupproted length of column about X (l_{ux}) =	1600mm
Unsupproted length of column about Z (l_{uz}) =	1600mm

For Columns not braced against sideway, Where, Effective length factor K= 2

Slenderness ratio,	K l_{ux} / r_x =	21.33	< 22 SECTION 6.2.5a	Radius of gyration r_x=	150	mm
	K l_{uz} / r_z =	21.33	< 22 SECTION 6.2.5a	Radius of gyration r_z=	150	mm

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γ_y = 0.76
γ_z = 0.76

Load Case	Load Combination	Factored Load From			Load at Column Base			Nominal Load/Moment			Kn = Pn/f_cbD (N/mm^2)
		Pu (kN)	Hux (kN)	Huz (kN)	Pu (kN)	Muz (kNm)	Mux (kNm)	Pn (kN)	Mnz (kNm)	Mnx (kNm)	
201	1.4(D+Cl)	26.01	3.1	0.0	35.61	4.9	0.0	54.8	7.5	0.0	0.006
202	1.4(D+Cl+Cr)	24.54	2.8	0.0	34.14	4.5	0.0	52.5	7.0	0.0	0.006
203	1.4(D+Cl+T1)+0.5Lr	29.23	4.2	0.0	38.83	6.7	0.0	59.7	10.3	0.0	0.007
204	1.4(D+Cl+T2)+0.5Lr	27.97	2.4	0.0	37.57	3.8	0.0	57.8	5.8	0.0	0.007
205	1.2(D+Cl)+1.6Lr+0.8WL1	10.67	-8.9	0.0	20.27	14.2	0.0	31.2	21.9	0.0	0.004
206	1.2(D+Cl)+1.6Lr+0.8WL2	15.00	-12.0	0.0	24.60	19.3	0.0	37.8	29.7	0.0	0.004
207	1.2(D+Cl)+1.6Lr+0.8WR1	27.76	12.9	0.0	37.36	20.7	0.0	57.5	31.8	0.0	0.007
208	1.2(D+Cl)+1.6Lr+0.8WR2	32.14	9.8	0.0	41.74	15.7	0.0	64.2	24.1	0.0	0.007
209	1.2(D+Cl)+1.6Lr+0.8W_Ln1	-7.94	8.0	-16.0	1.66	12.9	25.6	2.6	19.8	39.4	0.000
210	1.2(D+Cl)+1.6Lr+0.8W_Ln2	-3.56	4.9	-16.0	6.04	7.8	25.6	9.3	12.0	39.4	0.001
211	1.2(D+Cl)+1.6Lr+0.8W_Ln-1	-4.93	8.1	16.0	4.67	12.9	25.6	7.2	19.8	39.4	0.001
212	1.2(D+Cl)+1.6Lr+0.8W_Ln-2	-0.56	4.9	16.0	9.04	7.9	25.6	13.9	12.1	39.4	0.002
213	1.2(D+Cl)+0.5Lr+1.6WL1	-14.91	-21.6	0.0	-5.31	34.5	0.0	-8.2	53.1	0.0	-0.001
214	1.2(D+Cl)+0.5Lr+1.6WR1	-6.26	-27.9	0.0	3.35	44.6	0.0	5.1	68.7	0.0	0.001
215	1.2(D+Cl)+0.5Lr+1.6WL2	19.27	22.0	0.0	28.87	35.2	0.0	44.4	54.2	0.0	0.005
216	1.2(D+Cl)+0.5Lr+1.6WR2	28.03	15.8	0.0	37.63	25.2	0.0	57.9	38.8	0.0	0.007
217	1.2(D+Cl)+0.5Lr+1.6W_Ln1	-52.13	12.3	-32.0	-42.53	19.6	51.2	-65.4	30.2	78.8	-0.007
218	1.2(D+Cl)+0.5Lr+1.6W_Ln2	-43.38	6.0	-32.0	-33.78	9.6	51.2	-52.0	14.7	78.8	-0.006
219	1.2(D+Cl)+0.5Lr+1.6W_Ln-1	-46.11	12.3	32.0	-36.51	19.7	51.2	-56.2	30.3	78.8	-0.006
220	1.2(D+Cl)+0.5Lr+1.6W_Ln-2	-37.38	6.0	32.0	-27.78	9.7	51.2	-42.7	14.9	78.8	-0.005
221	1.2D+SL	1.82	0.1	10.0	11.42	0.1	16.0	17.6	0.2	24.6	0.002
222	1.2D+SR	17.90	0.9	-10.0	27.50	1.4	16.0	42.3	2.2	24.6	0.005
223	0.9D+1.6WL1	-32.39	-24.1	0.0	-22.79	38.5	0.0	-35.1	59.2	0.0	-0.004
224	0.9D+1.6WL2	-23.74	-30.4	0.0	-14.14	48.6	0.0	-21.8	74.8	0.0	-0.002
225	0.9D+1.6WR1	1.78	19.5	0.0	11.38	31.3	0.0	17.5	48.1	0.0	0.002
226	0.9D+1.6WR2	10.55	13.3	0.0	20.15	21.3	0.0	31.0	32.7	0.0	0.004
227	0.9D+1.6W_Ln1	-69.61	9.8	-32.0	-60.01	15.7	51.2	-92.3	24.1	78.8	-0.011
228	0.9D+1.6W_Ln2	-60.86	3.5	-32.0	-51.26	5.6	51.2	-78.9	8.6	78.8	-0.009
229	0.9D+1.6W_Ln-1	-63.59	9.8	32.0	-53.99	15.7	51.2	-83.1	24.2	78.8	-0.009
230	0.9D+1.6W_Ln-2	-54.86	3.6	32.0	-45.26	5.7	51.2	-69.6	8.8	78.8	-0.008
231	0.9D+1.0SL	-0.64	0.0	10.0	8.96	0.1	16.0	13.8	0.1	24.6	0.002
232	0.9D+1.0SR	15.44	0.8	-10.0	25.04	1.2	16.0	38.5	1.9	24.6	0.004

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Reinforcement: Provide,

4-Nos	20 mm Dia
2-Nos	20 mm Dia
2-Nos	20 mm Dia

1257.14 Sq mm (At Corner)
628.57 Sq mm (Along Z axis)
628.57 Sq mm (Along Y axis)

A_{sc} pvd. = 2514.29 Sq mm

Total A_{sc} = 1.01 % ρ_g = 0.010

β = P_n / P_0 (range is 0.55 to 0.70) Where, $P_0 = 0.85 f'_c (A_g - A_{sc}) + f_y A_{sc}$ = 841.87 T SECTION 22.4.2.2

Load Comb	From Chart of SP17(M)-R35-		M_{noz}	M_{nox}	β	$\alpha_n = \log(0.5)/\log(\beta)$	Interaction ratio	Remarks
	$R_{nz}=M_{noz}/f'_cb^2D$	$R_{nx}=M_{nox}/f'_cb^2D$						
1.4(D+Cl)	0.055	0.055	240.6	240.6	0.55	1.16	0.02	O.K
1.4(D+Cl+Cr)	0.055	0.055	240.6	240.6	0.55	1.16	0.02	O.K
1.4(D+Cl+T1)+0.5Lr	0.055	0.055	240.6	240.6	0.55	1.16	0.03	O.K
1.4(D+Cl+T2)+0.5Lr	0.055	0.055	240.6	240.6	0.55	1.16	0.01	O.K
1.2(D+Cl)+1.6Lr+0.8WL1	0.055	0.055	240.6	240.6	0.55	1.16	0.06	O.K
1.2(D+Cl)+1.6Lr+0.8WL2	0.055	0.055	240.6	240.6	0.55	1.16	0.09	O.K
1.2(D+Cl)+1.6Lr+0.8WR1	0.055	0.055	240.6	240.6	0.55	1.16	0.10	O.K
1.2(D+Cl)+1.6Lr+0.8WR2	0.055	0.055	240.6	240.6	0.55	1.16	0.07	O.K
1.2(D+Cl)+1.6Lr+0.8W_Ln1	0.055	0.055	240.6	240.6	0.55	1.16	0.18	O.K
1.2(D+Cl)+1.6Lr+0.8W_Ln2	0.055	0.055	240.6	240.6	0.55	1.16	0.15	O.K
1.2(D+Cl)+1.6Lr+0.8W_Ln-1	0.055	0.055	240.6	240.6	0.55	1.16	0.18	O.K
1.2(D+Cl)+1.6Lr+0.8W_Ln-2	0.055	0.055	240.6	240.6	0.55	1.16	0.15	O.K
1.2(D+Cl)+0.5Lr+1.6WL1	0.055	0.055	240.6	240.6	0.55	1.16	0.17	O.K
1.2(D+Cl)+0.5Lr+1.6WR1	0.055	0.055	240.6	240.6	0.55	1.16	0.23	O.K
1.2(D+Cl)+0.5Lr+1.6WL2	0.055	0.055	240.6	240.6	0.55	1.16	0.18	O.K
1.2(D+Cl)+0.5Lr+1.6WR2	0.055	0.055	240.6	240.6	0.55	1.16	0.12	O.K
1.2(D+Cl)+0.5Lr+1.6W_Ln1	0.055	0.055	240.6	240.6	0.55	1.16	0.36	O.K
1.2(D+Cl)+0.5Lr+1.6W_Ln2	0.055	0.055	240.6	240.6	0.55	1.16	0.31	O.K
1.2(D+Cl)+0.5Lr+1.6W_Ln-1	0.055	0.055	240.6	240.6	0.55	1.16	0.36	O.K
1.2(D+Cl)+0.5Lr+1.6W_Ln-2	0.055	0.055	240.6	240.6	0.55	1.16	0.31	O.K
1.2D+SL	0.055	0.055	240.6	240.6	0.55	1.16	0.07	O.K
1.2D+SR	0.055	0.055	240.6	240.6	0.55	1.16	0.08	O.K
0.9D+1.6WL1	0.055	0.055	240.6	240.6	0.55	1.16	0.20	O.K
0.9D+1.6WL2	0.055	0.055	240.6	240.6	0.55	1.16	0.26	O.K
0.9D+1.6WR1	0.055	0.055	240.6	240.6	0.55	1.16	0.15	O.K
0.9D+1.6WR2	0.055	0.055	240.6	240.6	0.55	1.16	0.10	O.K
0.9D+1.6W_Ln1	0.055	0.055	240.6	240.6	0.55	1.16	0.34	O.K
0.9D+1.6W_Ln2	0.055	0.055	240.6	240.6	0.55	1.16	0.29	O.K
0.9D+1.6W_Ln-1	0.055	0.055	240.6	240.6	0.55	1.16	0.34	O.K
0.9D+1.6W_Ln-2	0.055	0.055	240.6	240.6	0.55	1.16	0.30	O.K
0.9D+1.0SL	0.055	0.055	240.6	240.6	0.55	1.16	0.07	O.K
0.9D+1.0SR	0.055	0.055	240.6	240.6	0.55	1.16	0.07	O.K

Links, 10 mm dia @ 250 C/C

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5.3 Design calculation of Footing F2

FOUNDATION MKD. F2

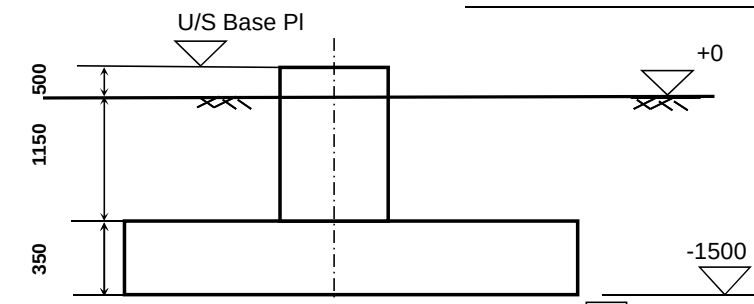
	Along-X	Along-Z
Size of column :	0.50m	0.50m
Size of fdn. raft :	1.80m	1.80m
Thickness of raft :	0.35 m	
Original ground level:	0.00 m	
Finished floor level :	0.00 m	
U/S level of foundation :	-1.50 m	
U/S level Base Plate :	0.50 m	
Unit weight of conc :	24 kN/m^3	
Unit weight of soil :	18 kN/m^3	

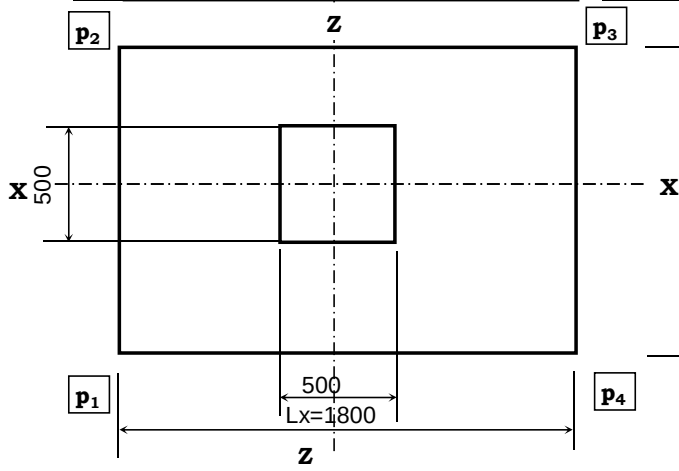
Calculation of Weight of Foundation :

Weight of raft & Col.= (1.8*1.8*0.35*24)+(0.5*0.5*1.65*24) = 37.12KN

Weight of soil = ((1.8*1.8)-(0.5*0.5))*1.15*18 = 61.89KN

Total = 99.01KN





Design Input:

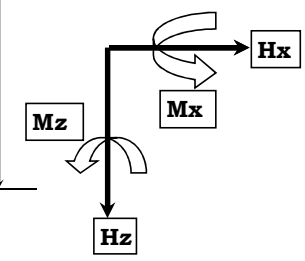
Characteristic strength of concrete, f '$_c$=	35 MPa
Yield stress of steel reinforcement, f_y=	420 MPa
Clear cover for main reinforcement bars =	75mm
Dia. of bars provd. along X direction at bottom =	12mm
Dia. of bars provided along Z direction at bottom=	12mm
Mininum percentage of reinforcement at bottom=	0.18%

Effective depth of slab along X direction = 269mm = 0.269m

Effective depth of slab along Z direction = 257mm = 0.257m

Clear projection along Z direction = 0.65m

Clear projection along X direction = 0.65m



H_X=Force along X-X direction

Hz=Force along Z-Z direction

M_X=Moment about X-X direction

Mz=Moment about Z-Z direction

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FOUNDATION MKD. F2

Sl. No.	Basic Load Case	V (kN)	Hx (kN)	Mz (kNm)	Hz (kN)	Mx (kNm)
1	Dead Load (D)	10.84	0.00	0.0	0.0	0.0
2	Collateral Load (Cl)	0.14	0.00	0.0	0.0	0.0
3	Roof Live (Lr)	14.18	0.00	0.0	0.0	0.0
4	Crane Load (Cr)	6.34	0.00	0.0	0.0	0.0
5	Wind_Left1 (WL1)	-41.01	0.00	0.0	-11.9	0.0
6	Wind_Left2 (WL2)	-29.59	0.00	0.0	-5.1	0.0
7	Wind_Right1 (WR1)	-30.35	0.00	0.0	11.9	0.0
8	Wind_Right2 (WR2)	-18.70	0.00	0.0	5.1	0.0
9	Wind_Long1 (W_Ln1)	-28.87	0.00	0.0	-11.9	0.0
10	Wind_Long2 (W_Ln2)	-17.29	0.00	0.0	-5.1	0.0
11	Wind_Long-1 (W_Ln-1)	-30.90	0.00	0.0	0.0	0.0
12	Wind_Long-2 (W_Ln-2)	-19.32	0.00	0.0	0.0	0.0
13	Seismic_L (SL)	-0.84	0.00	0.0	0.0	0.0
14	Seismic_R (SR)	0.84	0.00	0.0	0.0	0.0
15	Temparature 25 degree (T1)	-0.59	0.00	0.0	0.0	0.0
16	Temparature -25 degree (T2)	0.59	0.00	0.0	0.0	0.0

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Combined Load Cases:		Load at top of foundation					Load at bottom of foundation				
Load Case	Load Combination	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
Allowable Stress Design											
101	D+Cl	11.0	0.0	0.0	0.0	0.0	109.99	0.00	0.00	0.00	0.00
102	D+Cl+Lr+Cr	31.5	0.0	0.0	0.0	0.0	130.51	0.00	0.00	0.00	0.00
103	D+Cl+Cr+T1	16.7	0.0	0.0	0.0	0.0	115.74	0.00	0.00	0.00	0.00
104	D+Cl+Cr+T2	17.9	0.0	0.0	0.0	0.0	116.92	0.00	0.00	0.00	0.00
105	D+Cl+Cr+0.75Lr+0.75T1	27.5	0.0	0.0	0.0	0.0	126.52	0.00	0.00	0.00	0.00
106	D+Cl+Cr+0.75Lr+0.75T2	28.4	0.0	0.0	0.0	0.0	127.41	0.00	0.00	0.00	0.00
107	D+Cl+WL1	-30.0	0.0	0.0	-11.9	0.0	68.98	0.00	0.00	-11.88	-23.76
108	D+Cl+WL2	-18.6	0.0	0.0	-5.1	0.0	80.40	0.00	0.00	-5.09	-10.18
109	D+Cl+WR1	-19.4	0.0	0.0	11.9	0.0	79.64	0.00	0.00	11.88	23.76
110	D+Cl+WR2	-7.7	0.0	0.0	5.1	0.0	91.29	0.00	0.00	5.09	10.18
111	D+Cl+WLn1	-17.9	0.0	0.0	-11.9	0.0	81.12	0.00	0.00	-11.88	-23.76
112	D+Cl+W_Ln2	-6.3	0.0	0.0	-5.1	0.0	92.70	0.00	0.00	-5.09	-10.18
113	D+Cl+W_Ln-1	-19.9	0.0	0.0	0.0	0.0	79.09	0.00	0.00	0.00	0.00
114	D+Cl+W_Ln-2	-8.3	0.0	0.0	0.0	0.0	90.67	0.00	0.00	0.00	0.00
115	D+Cl+0.7SL	10.4	0.0	0.0	0.0	0.0	109.40	0.00	0.00	0.00	0.00
116	D+Cl+0.7SR	11.6	0.0	0.0	0.0	0.0	110.58	0.00	0.00	0.00	0.00
117	D+0.75Lr+0.75WL1	-9.3	0.0	0.0	-8.9	0.0	89.73	0.00	0.00	-8.91	-17.82
118	D+0.75Lr+0.75WL2	-0.7	0.0	0.0	-3.8	0.0	98.29	0.00	0.00	-3.82	-7.64
119	D+0.75Lr+0.75WR1	-1.3	0.0	0.0	8.9	0.0	97.72	0.00	0.00	8.91	17.82
120	D+0.75Lr+0.75WR2	7.5	0.0	0.0	3.8	0.0	106.46	0.00	0.00	3.82	7.64
121	D+0.75Lr+0.75W_Ln1	-0.2	0.0	0.0	-8.9	0.0	98.83	0.00	0.00	-8.91	-17.82
122	D+0.75Lr+0.75W_Ln2	8.5	0.0	0.0	-3.8	0.0	107.52	0.00	0.00	-3.82	-7.64
123	D+0.75Lr+0.75W_Ln-1	-1.7	0.0	0.0	0.0	0.0	97.31	0.00	0.00	0.00	0.00
124	D+0.75Lr+0.75W_Ln-2	7.0	0.0	0.0	0.0	0.0	106.00	0.00	0.00	0.00	0.00
125	D+0.75Lr+0.525SL	21.0	0.0	0.0	0.0	0.0	120.04	0.00	0.00	0.00	0.00
126	D+0.75Lr+0.525SR	21.9	0.0	0.0	0.0	0.0	120.93	0.00	0.00	0.00	0.00
127	0.6D+WL1	-34.5	0.0	0.0	-11.9	0.0	64.50	0.00	0.00	-11.88	-23.76
128	0.6D+WL2	-23.1	0.0	0.0	-5.1	0.0	75.92	0.00	0.00	-5.09	-10.18
129	0.6D+WR1	-23.8	0.0	0.0	11.9	0.0	75.16	0.00	0.00	11.88	23.76
130	0.6D+WR2	-12.2	0.0	0.0	5.1	0.0	86.81	0.00	0.00	5.09	10.18
131	0.6D+W_Ln1	-22.4	0.0	0.0	-11.9	0.0	76.64	0.00	0.00	-11.88	-23.76
132	0.6D+W_Ln2	-10.8	0.0	0.0	-5.1	0.0	88.22	0.00	0.00	-5.09	-10.18
133	0.6D+W_Ln-1	-24.4	0.0	0.0	0.0	0.0	74.61	0.00	0.00	0.00	0.00
134	0.6D+W_Ln-2	-12.8	0.0	0.0	0.0	0.0	86.19	0.00	0.00	0.00	0.00
135	0.6D+0.7SL	5.916	0	0	0	0	104.93	0.00	0.00	0.00	0.00
136	0.6D+0.7SR	7.092	0	0	0	0	106.10	0.00	0.00	0.00	0.00

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Ultimate Conditions											
201	1.4(D+Cl)	15.4	0.0	0.0	0.0	0.0	114.38	0.00	0.00	0.00	0.00
202	1.4(D+Cl+Cr)	24.2	0.0	0.0	0.0	0.0	123.26	0.00	0.00	0.00	0.00
203	1.4(D+Cl+T1)+0.5Lr	21.6	0.0	0.0	0.0	0.0	120.65	0.00	0.00	0.00	0.00
204	1.4(D+Cl+T2)+0.5Lr	23.3	0.0	0.0	0.0	0.0	122.30	0.00	0.00	0.00	0.00
205	1.2(D+Cl)+1.6Lr+0.8WL1	3.1	0.0	0.0	-9.5	0.0	102.07	0.00	0.00	-9.50	-19.01
206	1.2(D+Cl)+1.6Lr+0.8WL2	12.2	0.0	0.0	-4.1	0.0	111.20	0.00	0.00	-4.07	-8.14
207	1.2(D+Cl)+1.6Lr+0.8WR1	11.6	0.0	0.0	9.5	0.0	110.59	0.00	0.00	9.50	19.01
208	1.2(D+Cl)+1.6Lr+0.8WR2	20.9	0.0	0.0	4.1	0.0	119.91	0.00	0.00	4.07	8.14
209	1.2(D+Cl)+1.6Lr+0.8W_Ln1	12.8	0.0	0.0	-9.5	0.0	111.78	0.00	0.00	-9.50	-19.01
210	1.2(D+Cl)+1.6Lr+0.8W_Ln2	22.0	0.0	0.0	-4.1	0.0	121.04	0.00	0.00	-4.07	-8.14
211	1.2(D+Cl)+1.6Lr+0.8W_Ln-1	11.1	0.0	0.0	0.0	0.0	110.15	0.00	0.00	0.00	0.00
212	1.2(D+Cl)+1.6Lr+0.8W_Ln-2	20.4	0.0	0.0	0.0	0.0	119.42	0.00	0.00	0.00	0.00
213	1.2(D+Cl)+0.5Lr+1.6WL1	-45.35	0.00	0	-19.008	0	53.66	0.00	0.00	-19.01	-38.02
214	1.2(D+Cl)+0.5Lr+1.6WR1	-27.08	0.00	0	-8.144	0	71.93	0.00	0.00	-8.14	-16.29
215	1.2(D+Cl)+0.5Lr+1.6WL2	-28.29	0.00	0	19.008	0	70.72	0.00	0.00	19.01	38.02
216	1.2(D+Cl)+0.5Lr+1.6WR2	-9.65	0.00	0	8.144	0	89.36	0.00	0.00	8.14	16.29
217	1.2(D+Cl)+0.5Lr+1.6W_Ln1	-25.93	0.00	0	-19.008	0	73.08	0.00	0.00	-19.01	-38.02
218	1.2(D+Cl)+0.5Lr+1.6W_Ln2	-7.40	0.00	0	-8.144	0	91.61	0.00	0.00	-8.14	-16.29
219	1.2(D+Cl)+0.5Lr+1.6W_Ln-1	-29.17	0.00	0	0	0	69.84	0.00	0.00	0.00	0.00
220	1.2(D+Cl)+0.5Lr+1.6W_Ln-2	-10.65	0.00	0	0	0	88.36	0.00	0.00	0.00	0.00
221	1.2D+SL	12.17	0.00	0	0	0	111.18	0.00	0.00	0.00	0.00
222	1.2D+SR	13.85	0.00	0	0	0	112.86	0.00	0.00	0.00	0.00
223	0.9D+1.6WL1	-55.86	0.00	0	-19.008	0	43.15	0.00	0.00	-19.01	-38.02
224	0.9D+1.6WL2	-37.59	0.00	0	-8.144	0	61.42	0.00	0.00	-8.14	-16.29
225	0.9D+1.6WR1	-38.80	0.00	0	19.008	0	60.21	0.00	0.00	19.01	38.02
226	0.9D+1.6WR2	-20.16	0.00	0	8.144	0	78.85	0.00	0.00	8.14	16.29
227	0.9D+1.6W_Ln1	-36.44	0.00	0	-19.008	0	62.57	0.00	0.00	-19.01	-38.02
228	0.9D+1.6W_Ln2	-17.91	0.00	0	-8.144	0	81.10	0.00	0.00	-8.14	-16.29
229	0.9D+1.6W_Ln-1	-39.68	0.00	0	0	0	59.33	0.00	0.00	0.00	0.00
230	0.9D+1.6W_Ln-2	-21.16	0.00	0	0	0	77.85	0.00	0.00	0.00	0.00
231	0.9D+1.0SL	8.92	0.00	0	0	0	107.93	0.00	0.00	0.00	0.00
232	0.9D+1.0SR	10.60	0.00	0	0	0	109.61	0.00	0.00	0.00	0.00

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Check for overturning and sliding:

	Load Cases:	Load at b/o fdn					F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
		V (kN)	Hx (kN)	Mz (kNm)	Hz (kN)	Mx (kNm)				
101	D+Cl	109.99	0.00	0.00	0.00	0.00	NA	NA	NA	NA
102	D+Cl+Lr+Cr	130.51	0.00	0.00	0.00	1.00	NA	117	NA	NA
103	D+Cl+Cr+T1	115.74	0.00	0.00	0.00	2.00	NA	52	NA	NA
104	D+Cl+Cr+T2	116.92	0.00	0.00	0.00	3.00	NA	35	NA	NA
105	D+Cl+Cr+0.75Lr+0.75T1	126.52	0.00	0.00	0.00	4.00	NA	28	NA	NA
106	D+Cl+Cr+0.75Lr+0.75T2	127.41	0.00	0.00	0.00	5.00	NA	23	NA	NA
107	D+Cl+WL1	68.98	0.00	0.00	-11.88	6.00	NA	10	NA	2.32
108	D+Cl+WL2	80.40	0.00	0.00	-5.09	7.00	NA	10	NA	6.32
109	D+Cl+WR1	79.64	0.00	0.00	11.88	8.00	NA	9	NA	2.68
110	D+Cl+WR2	91.29	0.00	0.00	5.09	9.00	NA	9	NA	7.17
111	D+Cl+WLn1	81.12	0.00	0.00	-11.88	10.00	NA	7	NA	2.73
112	D+Cl+W_Ln2	92.70	0.00	0.00	-5.09	11.00	NA	8	NA	7.28
113	D+Cl+W_Ln-1	79.09	0.00	0.00	0.00	12.00	NA	6	NA	NA
114	D+Cl+W_Ln-2	90.67	0.00	0.00	0.00	13.00	NA	6	NA	NA
115	D+Cl+0.7SL	109.40	0.00	0.00	0.00	14.00	NA	7	NA	NA
116	D+Cl+0.7SR	110.58	0.00	0.00	0.00	15.00	NA	7	NA	NA
117	D+0.75Lr+0.75WL1	89.73	0.00	0.00	-8.91	16.00	NA	5	NA	4.03
118	D+0.75Lr+0.75WL2	98.29	0.00	0.00	-3.82	17.00	NA	5	NA	10.30
119	D+0.75Lr+0.75WR1	97.72	0.00	0.00	8.91	18.00	NA	5	NA	4.39
120	D+0.75Lr+0.75WR2	106.46	0.00	0.00	3.82	19.00	NA	5	NA	11.15
121	D+0.75Lr+0.75W_Ln1	98.83	0.00	0.00	-8.91	20.00	NA	4	NA	4.44
122	D+0.75Lr+0.75W_Ln2	107.52	0.00	0.00	-3.82	21.00	NA	5	NA	11.27
123	D+0.75Lr+0.75W_Ln-1	97.31	0.00	0.00	0.00	22.00	NA	4	NA	NA
124	D+0.75Lr+0.75W_Ln-2	106.00	0.00	0.00	0.00	23.00	NA	4	NA	NA
125	D+0.75Lr+0.525SL	120.04	0.00	0.00	0.00	24.00	NA	5	NA	NA
126	D+0.75Lr+0.525SR	120.93	0.00	0.00	0.00	25.00	NA	4	NA	NA
127	0.6D+WL1	64.50	0.00	0.00	-11.88	26.00	NA	2	NA	2.17
128	0.6D+WL2	75.92	0.00	0.00	-5.09	27.00	NA	3	NA	5.97
129	0.6D+WR1	75.16	0.00	0.00	11.88	28.00	NA	2	NA	2.53
130	0.6D+WR2	86.81	0.00	0.00	5.09	29.00	NA	3	NA	6.82
131	0.6D+W_Ln1	76.64	0.00	0.00	-11.88	30.00	NA	2	NA	2.58
132	0.6D+W_Ln2	88.22	0.00	0.00	-5.09	31.00	NA	3	NA	6.93
133	0.6D+W_Ln-1	74.61	0.00	0.00	0.00	32.00	NA	2	NA	NA
134	0.6D+W_Ln-2	86.19	0.00	0.00	0.00	33.00	NA	2	NA	NA
135	0.6D+0.7SL	104.93	0.00	0.00	0.00	34.00	NA	3	NA	NA
136	0.6D+0.7SR	106.10	0.00	0.00	0.00	35.00	NA	3	NA	NA

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FOUNDATION MKD. F2

Load Combination & Calculation of Base pressure

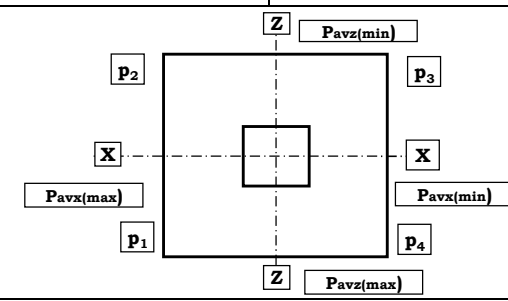
Max. Base pressure = $p_{max} = (V/(Lx.Lz)) + (6.Mz/((Lx)^2.Lz) + (6.Mx/(Lx.(Lz)^2)$

Min. Base pressure = $p_{min} = (V/(Lx.Lz)) - (6.Mz/((Lx)^2.Lz) - (6.Mx/(Lx.(Lz)^2)$

Surcharge load = 20 kN/m^2

Net SBC of soil for DL, LL= 400 kN/m^2 For WL/SL case = 500 kN/m^2

Gross SBC of soil for DL, LL= 427 kN/m^2 For WL/SL case = 527 kN/m^2



Load Case	Load Combination	Design Load/moments			Base Pressure (kN/m^2)												
		V (kN)	Mz (kN-m)	Mx (kN-m)	V/ (Lx.Lz)	6.Mz/ (Lx)^2.Lz	6.Mx/ Lx.(Lz)^2	p_1 (p_{max})	p_2	p_4	p_3 (p_{min})	(p_{max}-γh)	$p_{avx(max)}$	$p_{avx(min)}$	$p_{avz(max)}$	$p_{avz(min)}$	Remarks
101	D+Cl	109.99	0.00	0.00	33.95	0.00	0.00	33.95	33.95	33.95	33.95	6.95	33.95	33.95	33.95	33.95	O.K
102	D+Cl+Lr+Cr	130.51	0.00	0.00	40.28	0.00	0.00	40.28	40.28	40.28	40.28	13.28	40.28	40.28	40.28	40.28	O.K
103	D+Cl+Cr+T1	115.74	0.00	0.00	35.72	0.00	0.00	35.72	35.72	35.72	35.72	8.72	35.72	35.72	35.72	35.72	O.K
104	D+Cl+Cr+T2	116.92	0.00	0.00	36.09	0.00	0.00	36.09	36.09	36.09	36.09	9.09	36.09	36.09	36.09	36.09	O.K
105	D+Cl+Cr+0.75Lr+0.75T1	126.52	0.00	0.00	39.05	0.00	0.00	39.05	39.05	39.05	39.05	12.05	39.05	39.05	39.05	39.05	O.K
106	D+Cl+Cr+0.75Lr+0.75T2	127.41	0.00	0.00	39.32	0.00	0.00	39.32	39.32	39.32	39.32	12.32	39.32	39.32	39.32	39.32	O.K
107	D+Cl+WL1	68.98	0.00	23.76	21.29	0.00	24.44	45.73	-3.15	45.73	-3.15	18.73	21.29	21.29	45.73	-3.15	O.K
108	D+Cl+WL2	80.40	0.00	10.18	24.81	0.00	10.47	35.29	14.34	35.29	14.34	8.29	24.82	24.82	35.29	14.34	O.K
109	D+Cl+WR1	79.64	0.00	23.76	24.58	0.00	24.44	49.02	0.14	49.02	0.14	22.02	24.58	24.58	49.02	0.14	O.K
110	D+Cl+WR2	91.29	0.00	10.18	28.18	0.00	10.47	38.65	17.70	38.65	17.70	11.65	28.18	28.18	38.65	17.70	O.K
111	D+Cl+WLn1	81.12	0.00	23.76	25.04	0.00	24.44	49.48	0.59	49.48	0.59	22.48	25.04	25.04	49.48	0.59	O.K
112	D+Cl+W_Ln2	92.70	0.00	10.18	28.61	0.00	10.47	39.08	18.14	39.08	18.14	12.08	28.61	28.61	39.08	18.14	O.K
113	D+Cl+W_Ln-1	79.09	0.00	0.00	24.41	0.00	0.00	24.41	24.41	24.41	24.41	-2.59	24.41	24.41	24.41	24.41	O.K
114	D+Cl+W_Ln-2	90.67	0.00	0.00	27.98	0.00	0.00	27.98	27.98	27.98	27.98	0.98	27.98	27.98	27.98	27.98	O.K
115	D+Cl+0.7SL	109.40	0.00	0.00	33.77	0.00	0.00	33.77	33.77	33.77	33.77	6.77	33.77	33.77	33.77	33.77	O.K
116	D+Cl+0.7SR	110.58	0.00	0.00	34.13	0.00	0.00	34.13	34.13	34.13	34.13	7.13	34.13	34.13	34.13	34.13	O.K
117	D+0.75Lr+0.75WL1	89.73	0.00	17.82	27.69	0.00	18.33	46.03	9.36	46.03	9.36	19.03	27.70	27.70	46.03	9.36	O.K
118	D+0.75Lr+0.75WL2	98.29	0.00	7.64	30.34	0.00	7.85	38.19	22.48	38.19	22.48	11.19	30.34	30.34	38.19	22.48	O.K
119	D+0.75Lr+0.75WR1	97.72	0.00	17.82	30.16	0.00	18.33	48.49	11.83	48.49	11.83	21.49	30.16	30.16	48.49	11.83	O.K
120	D+0.75Lr+0.75WR2	106.46	0.00	7.64	32.86	0.00	7.85	40.71	25.00	40.71	25.00	13.71	32.86	32.86	40.71	25.00	O.K
121	D+0.75Lr+0.75W_Ln1	98.83	0.00	17.82	30.50	0.00	18.33	48.84	12.17	48.84	12.17	21.84	30.51	30.51	48.84	12.17	O.K
122	D+0.75Lr+0.75W_Ln2	107.52	0.00	7.64	33.18	0.00	7.85	41.04	25.33	41.04	25.33	14.04	33.19	33.19	41.04	25.33	O.K
123	D+0.75Lr+0.75W_Ln-1	97.31	0.00	0.00	30.03	0.00	0.00	30.03	30.03	30.03	30.03	3.03	30.03	30.03	30.03	30.03	O.K
124	D+0.75Lr+0.75W_Ln-2	106.00	0.00	0.00	32.71	0.00	0.00	32.71	32.71	32.71	32.71	5.71	32.71	32.71	32.71	32.71	O.K
125	D+0.75Lr+0.525SL	120.04	0.00	0.00	37.05	0.00	0.00	37.05	37.05	37.05	37.05	10.05	37.05	37.05	37.05	37.05	O.K
126	D+0.75Lr+0.525SR	120.93	0.00	0.00	37.32	0.00	0.00	37.32	37.32	37.32	37.32	10.32	37.32	37.32	37.32	37.32	O.K
127	0.6D+WL1	64.50	0.00	23.76	19.91	0.00	24.44	44.35	-4.54	44.35	-4.54	17.35	19.91	19.91	44.35	-4.54	O.K
128	0.6D+WL2	75.92	0.00	10.18	23.43	0.00	10.47	33.91	12.96	33.91	12.96	6.91	23.44	23.44	33.91	12.96	O.K
129	0.6D+WR1	75.16	0.00	23.76	23.20	0.00	24.44	47.64	-1.25	47.64	-1.25	20.64	23.20	23.20	47.64	-1.25	O.K
130	0.6D+WR2	86.81	0.00	10.18	26.79	0.00	10.47	37.27	16.32	37.27	16.32	10.27	26.80	26.80	37.27	16.32	O.K
131	0.6D+W_Ln1	76.64	0.00	23.76	23.66	0.00	24.44	48.10	-0.79	48.10	-0.79	21.10	23.66	23.66	48.10	-0.79	O.K
132	0.6D+W_Ln2	88.22	0.00	10.18	27.23	0.00	10.47	37.70	16.76	37.70	16.76	10.70	27.23	27.23	37.70	16.76	O.K
133	0.6D+W_Ln-1	74.61	0.00	0.00	23.03	0.00	0.00	23.03	23.03	23.03	23.03	-3.97	23.03	23.03	23.03	23.03	O.K
134	0.6D+W_Ln-2	86.19	0.00	0.00	26.60	0.00	0.00	26.60	26.60	26.60	26.60	-0.40	26.60	26.60	26.60	26.60	O.K
135	0.6D+0.7SL	104.93	0.00	0.00	32.38	0.00	0.00	32.38	32.38	32.38	32.38	5.38	32.38	32.38	32.38	32.38	O.K
136	0.6D+0.7SR	106.10	0.00	0.00	32.75	0.00	0.00	32.75	32.75	32.75	32.75	5.75	32.75	32.75	32.75	32.75	O.K

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201	1.4(D+Cl)	114.38	0.00	0.00	35.30	0.00	0.00	35.30	35.30	35.30	35.30	8.30	35.30	35.30	35.30	35.30
202	1.4(D+Cl+Cr)	123.26	0.00	0.00	38.04	0.00	0.00	38.04	38.04	38.04	38.04	11.04	38.04	38.04	38.04	38.04
203	1.4(D+Cl+T1)+0.5Lr	120.65	0.00	0.00	37.24	0.00	0.00	37.24	37.24	37.24	37.24	10.24	37.24	37.24	37.24	37.24
204	1.4(D+Cl+T2)+0.5Lr	122.30	0.00	0.00	37.75	0.00	0.00	37.75	37.75	37.75	37.75	10.75	37.75	37.75	37.75	37.75
205	1.2(D+Cl)+1.6Lr+0.8WL1	102.07	0.00	19.01	31.50	0.00	19.56	51.06	11.95	51.06	11.95	24.06	31.51	31.51	51.06	11.95
206	1.2(D+Cl)+1.6Lr+0.8WL2	111.20	0.00	8.14	34.32	0.00	8.38	42.70	25.94	42.70	25.94	15.70	34.32	34.32	42.70	25.94
207	1.2(D+Cl)+1.6Lr+0.8WR1	110.59	0.00	19.01	34.13	0.00	19.56	53.69	14.58	53.69	14.58	26.69	34.14	34.14	53.69	14.58
208	1.2(D+Cl)+1.6Lr+0.8WR2	119.91	0.00	8.14	37.01	0.00	8.38	45.39	28.63	45.39	28.63	18.39	37.01	37.01	45.39	28.63
209	1.2(D+Cl)+1.6Lr+0.8W_L	111.78	0.00	19.01	34.50	0.00	19.56	54.05	14.94	54.05	14.94	27.05	34.50	34.50	54.05	14.94
210	1.2(D+Cl)+1.6Lr+0.8W_L	121.04	0.00	8.14	37.36	0.00	8.38	45.74	28.98	45.74	28.98	18.74	37.36	37.36	45.74	28.98
211	1.2(D+Cl)+1.6Lr+0.8W_L	110.15	0.00	0.00	34.00	0.00	0.00	34.00	34.00	34.00	34.00	7.00	34.00	34.00	34.00	34.00
212	1.2(D+Cl)+1.6Lr+0.8W_L	119.42	0.00	0.00	36.86	0.00	0.00	36.86	36.86	36.86	36.86	9.86	36.86	36.86	36.86	36.86
213	1.2(D+Cl)+0.5Lr+1.6WL1	53.66	0.00	38.02	16.56	0.00	39.11	55.67	-22.55	55.67	-22.55	28.67	16.56	16.56	55.67	-22.55
214	1.2(D+Cl)+0.5Lr+1.6WR1	71.93	0.00	16.29	22.20	0.00	16.76	38.96	5.44	38.96	5.44	11.96	22.20	22.20	38.96	5.44
215	1.2(D+Cl)+0.5Lr+1.6WL2	70.72	0.00	38.02	21.83	0.00	39.11	60.94	-17.29	60.94	-17.29	33.94	21.83	21.83	60.94	-17.29
216	1.2(D+Cl)+0.5Lr+1.6WR2	89.36	0.00	16.29	27.58	0.00	16.76	44.34	10.82	44.34	10.82	17.34	27.58	27.58	44.34	10.82
217	1.2(D+Cl)+0.5Lr+1.6W_L	73.08	0.00	38.02	22.56	0.00	39.11	61.67	-16.55	61.67	-16.55	34.67	22.56	22.56	61.67	-16.55
218	1.2(D+Cl)+0.5Lr+1.6W_L	91.61	0.00	16.29	28.28	0.00	16.76	45.03	11.52	45.03	11.52	18.03	28.28	28.28	45.03	11.52
219	1.2(D+Cl)+0.5Lr+1.6W_L	69.84	0.00	0.00	21.55	0.00	0.00	21.55	21.55	21.55	21.55	-5.45	21.55	21.55	21.55	21.55
220	1.2(D+Cl)+0.5Lr+1.6W_L	88.36	0.00	0.00	27.27	0.00	0.00	27.27	27.27	27.27	27.27	0.27	27.27	27.27	27.27	27.27
221	1.2D+SL	111.18	0.00	0.00	34.31	0.00	0.00	34.31	34.31	34.31	34.31	7.31	34.31	34.31	34.31	34.31
222	1.2D+SR	112.86	0.00	0.00	34.83	0.00	0.00	34.83	34.83	34.83	34.83	7.83	34.83	34.83	34.83	34.83
223	0.9D+1.6WL1	43.15	0.00	38.02	13.32	0.00	39.11	52.43	-25.79	52.43	-25.79	25.43	13.32	13.32	52.43	-25.79
224	0.9D+1.6WL2	61.42	0.00	16.29	18.96	0.00	16.76	35.71	2.20	35.71	2.20	8.71	18.96	18.96	35.71	2.20
225	0.9D+1.6WR1	60.21	0.00	38.02	18.58	0.00	39.11	57.69	-20.53	57.69	-20.53	30.69	18.58	18.58	57.69	-20.53
226	0.9D+1.6WR2	78.85	0.00	16.29	24.34	0.00	16.76	41.09	7.58	41.09	7.58	14.09	24.34	24.34	41.09	7.58
227	0.9D+1.6W_Ln1	62.57	0.00	38.02	19.31	0.00	39.11	58.42	-19.80	58.42	-19.80	31.42	19.31	19.31	58.42	-19.80
228	0.9D+1.6W_Ln2	81.10	0.00	16.29	25.03	0.00	16.76	41.79	8.27	41.79	8.27	14.79	25.03	25.03	41.79	8.27
229	0.9D+1.6W_Ln-1	59.33	0.00	0.00	18.31	0.00	0.00	18.31	18.31	18.31	18.31	-8.69	18.31	18.31	18.31	18.31
230	0.9D+1.6W_Ln-2	77.85	0.00	0.00	24.03	0.00	0.00	24.03	24.03	24.03	24.03	-2.97	24.03	24.03	24.03	24.03
231	0.9D+1.0SL	107.93	0.00	0.00	33.31	0.00	0.00	33.31	33.31	33.31	33.31	6.31	33.31	33.31	33.31	33.31
232	0.9D+1.0SR	109.61	0.00	0.00	33.83	0.00	0.00	33.83	33.83	33.83	33.83	6.83	33.83	33.83	33.83	33.83
	Max :	131	0	38												
	Min :	43	0	0												

Max pressure at base of foundation without surcharge excluding weight of soil (DL+LL cases) = 13.28 kN/m^2

Max pressure at base of foundation including surcharge excluding weight of soil (DL+LL cases) = 33.28 kN/m^2

Max pressure at base of foundation without surcharge excluding weight of soil (DL+LL+SL/WL cases) = 34.67 kN/m^2

Max pressure at base of foundation including surcharge excluding weight of soil (DL+LL+WL/SL cases) = 54.67 kN/m^2

Sample calculation of values of base pressure:

Load case : D+Cl+Lr+Cr

Maximum pressure, p_1 (pmax) = 40.28+0.00+0.00= 40.28 kN/m^2

p_2= 40.28+0.00-0.00= 40.28 kN/m^2

p_3= 40.28-(0.00+0.00)= 40.28 kN/m^2

p_4= 40.28-0.00+0.00= 40.28 kN/m^2

Maxm pressure excluding weight of soil (p_{max}-γh)= 40.28-1.5*18= 13.28 kN/m^2

Maximum average Pressure along X, $p_{avx(max)}$ = (40.28+40.28)/2= 40.28 kN/m^2

Minimum average Pressure along X, $p_{avx(minx)}$ = (40.28+40.28)/2= 40.28 kN/m^2

Maximum average Pressure along Y, $p_{avy(max)}$ = (40.28+40.28)/2= 40.28 kN/m^2

Minimum average Pressure along Y, $p_{avy(minx)}$ = (40.28+40.28)/2= 40.28 kN/m^2

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FOUNDATION MKD. F2

Calculation of Bending Moment and Shear Force in X direction :

p_{1x} =pressure at the face of column at section 1x-1x in kN/m^2.

p_{2x} =pressure at the face of column at section 2x-2x in kN/m^2.

p_{3x} =pressure at a distance 'd' from the face of column at section 3x-3x in kN/m^2.

BM_{1x} =Moment at the face of column at section 1x-1x in kN-m/m.

BM_{2x} =Moment at the face of column 2x-2x in kN-m/m.

SF_{3x} = Oneway shear force at a distance 'd' from the face of column at section 3x-3x in kN/m.

SF (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Punching perimeter= ((0.5+2*0.5*0.257)+(0.5+2*0.5*0.257))*2

= 3.028 m

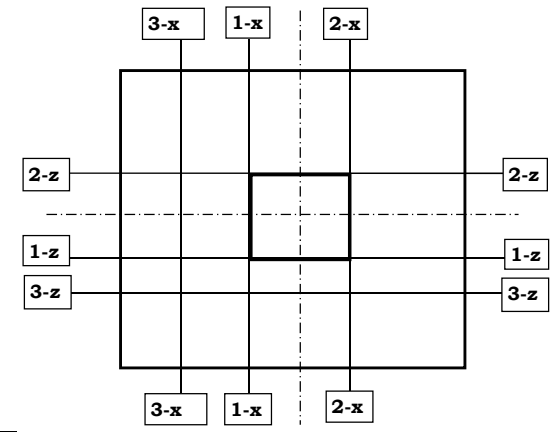
Punching Area= ((0.5+2*0.5*0.257)*(0.5+2*0.5*0.257))

= 0.57 m^2 Punching shear stress to be checked

Downward pressure due to self weight of raft and soil = (0.35*24+1.15*18)

= 29.10 kN/m^2

Area of footing, A_f= 3.24 m^2



Load Case	Load Combination	p_{1x} kN/m2	p_{2x} kN/m2	p_{3x} kN/m2	$B.M_{1x}$ kNm	$B.M_{2x}$ kNm	$S.F_{3x}$ kN	$S.F_{(punching)}$ kN
Ultimate Condition								
201	1.4(D+Cl)	35.30	35.30	35.30	1.31	1.31	2.36	16.56
202	1.4(D+Cl+Cr)	38.04	38.04	38.04	1.89	1.89	3.41	23.88
203	1.4(D+Cl+T1)+0.5Lr	37.24	37.24	37.24	1.72	1.72	3.10	21.72
204	1.4(D+Cl+T2)+0.5Lr	37.75	37.75	37.75	1.83	1.83	3.30	23.09
205	1.2(D+Cl)+1.6Lr+0.8WL1	31.51	31.51	31.51	0.51	0.51	0.92	6.41
206	1.2(D+Cl)+1.6Lr+0.8WL2	34.32	34.32	34.32	1.10	1.10	1.99	13.94
207	1.2(D+Cl)+1.6Lr+0.8WR1	34.14	34.14	34.14	1.06	1.06	1.92	13.44
208	1.2(D+Cl)+1.6Lr+0.8WR2	37.01	37.01	37.01	1.67	1.67	3.01	21.12
209	1.2(D+Cl)+1.6Lr+0.8W_Ln1	34.50	34.50	34.50	1.14	1.14	2.06	14.42
210	1.2(D+Cl)+1.6Lr+0.8W_Ln2	37.36	37.36	37.36	1.74	1.74	3.15	22.05
211	1.2(D+Cl)+1.6Lr+0.8W_Ln-	34.00	34.00	34.00	1.04	1.04	1.87	13.08
212	1.2(D+Cl)+1.6Lr+0.8W_Ln-	36.86	36.86	36.86	1.64	1.64	2.96	20.71
213	1.2(D+Cl)+0.5Lr+1.6WL1	16.56	16.56	16.56	-2.65	-2.65	-4.78	-33.48
214	1.2(D+Cl)+0.5Lr+1.6WR1	22.20	22.20	22.20	-1.46	-1.46	-2.63	-18.42
215	1.2(D+Cl)+0.5Lr+1.6WL2	21.83	21.83	21.83	-1.54	-1.54	-2.77	-19.42
216	1.2(D+Cl)+0.5Lr+1.6WR2	27.58	27.58	27.58	-0.32	-0.32	-0.58	-4.06
217	1.2(D+Cl)+0.5Lr+1.6W_Ln1	22.56	22.56	22.56	-1.38	-1.38	-2.49	-17.47
218	1.2(D+Cl)+0.5Lr+1.6W_Ln2	28.28	28.28	28.28	-0.17	-0.17	-0.31	-2.20
219	1.2(D+Cl)+0.5Lr+1.6W_Ln-	21.55	21.55	21.55	-1.59	-1.59	-2.88	-20.15
220	1.2(D+Cl)+0.5Lr+1.6W_Ln-	27.27	27.27	27.27	-0.39	-0.39	-0.70	-4.88
221	1.2D+SL	34.31	34.31	34.31	1.10	1.10	1.99	13.92
222	1.2D+SR	34.83	34.83	34.83	1.21	1.21	2.18	15.31
223	0.9D+1.6WL1	13.32	13.32	13.32	-3.33	-3.33	-6.01	-42.14
224	0.9D+1.6WL2	18.96	18.96	18.96	-2.14	-2.14	-3.86	-27.08
225	0.9D+1.6WR1	18.58	18.58	18.58	-2.22	-2.22	-4.01	-28.08
226	0.9D+1.6WR2	24.34	24.34	24.34	-1.01	-1.01	-1.81	-12.72
227	0.9D+1.6W_Ln1	19.31	19.31	19.31	-2.07	-2.07	-3.73	-26.13
228	0.9D+1.6W_Ln2	25.03	25.03	25.03	-0.86	-0.86	-1.55	-10.86
229	0.9D+1.6W_Ln-1	18.31	18.31	18.31	-2.28	-2.28	-4.11	-28.81
230	0.9D+1.6W_Ln-2	24.03	24.03	24.03	-1.07	-1.07	-1.93	-13.54
231	0.9D+1.0SL	33.31	33.31	33.31	0.89	0.89	1.60	11.24
232	0.9D+1.0SR	33.83	33.83	33.83	1.00	1.00	1.80	12.63

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Sample calculation of BM & SF

Load case : 201

p_{1x} =	35.3-(35.3-35.3)*0.65/1.8 =	35.30 kN/m²	Overhang =	0.65	m
p_{2x} =	35.3+(35.3-35.3)*0.65/1.8 =	35.30 kN/m²	Effective depth =	0.269	m
p_{3x} =	35.3-(35.3-35.3)*(0.65-0.269)/1.8 =	35.30 kN/m²			

BM_{1x} =	(35.30-29.1)*0.65*0.65/2+0.5*(35.3-35.30)*0.65*0.65*0.67 =	1.31	kN-m/m
BM_{2x} =	(35.30-29.1)*0.65*0.65/2-0.5*(-35.3+35.30)*0.65*0.65*0.67 =	1.31	kN-m/m
SF_{3x} =	(35.30-29.1)*(0.65-0.269)+0.5*(35.3-35.30)*(0.65-0.269) =	2.36	kN/m
$S.F_{(punching)}$ =	(3.24-0.57)*(35.30-29.1) =	16.56	kN

Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	1.89	0.03	0.03	0.000	22.30	484.20	12ϕT	233.58	300	200	565.49
At top	-3.33	-0.05	0.06	0.000	39.38	484.20	12ϕT	233.58	300	200	565.49

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, $(\phi\ Vc)/bd$ = 0.740 N/mm2

Shear at a distance 'd' from face of column, Vu= 3.41 kN/m

Shear stress , tv= (3.41*1000)/(1000*269)

= 0.01 N/mm2 **O.K.**

Check against Two-way shear force at a distance d/2 from face of column:

Design punching shear at a distance d/2 from face of column, Vu=	23.88 kN	
Strength of Concrete in shear Vc1 =	2409.42 kN	Table 22.6.5.2_ (a)
Strength of Concrete in shear Vc2 =	1516.63 kN	Table 22.6.5.2_ (b)
Strength of Concrete in shear Vc3 =	1606.28 kN	Table 22.6.5.2_ (c)
Design Vc=	1516.63 kN	
Allowable shear stress, $(\phi\ Vc)/bd$ =	1.40 N/mm2	

Design punching shear stress, τ = (23.88*1000)/(3028*269)

= 0.03 N/mm^2 **No shear reinforcement reqd.**

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Calculation of Bending Moment and Shear Force in Z direction :

p_{1z} =pressure at the face of column at section 1z-1z in kN/m^2.
p_{2z} =pressure at the face of column at section 2z-2z in kN/m^2.
p_{3z} =pressure at a distance d from the face of column at section 3z-3z in kN/m^2.
$B.M_{1z}$ =Moment at the face of column at section 1z-1z in kN-m/m.
$B.M_{2z}$ =Moment at the face of column at section 2z-2z in kN-m/m.
$S.F_{3z}$ = Oneway Shear Force at a distance d from the face of column at section 3z-3z in kN/m.
S.F (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Load Case	Load Combination	p_{1z} kN/m2	p_{2z} kN/m2	p_{3z} kN/m2	$B.M_{1z}$ kNm	$B.M_{2z}$ kNm	$S.F_{3z}$ kN
Ultimate Conditions							
201	1.4(D+Cl)	35.30	35.30	35.30	1.31	1.31	2.44
202	1.4(D+Cl+Cr)	38.04	38.04	38.04	1.89	1.89	3.51
203	1.4(D+Cl+T1)+0.5Lr	37.24	37.24	37.24	1.72	1.72	3.20
204	1.4(D+Cl+T2)+0.5Lr	37.75	37.75	37.75	1.83	1.83	3.40
205	1.2(D+Cl)+1.6Lr+0.8WL1	36.94	26.07	42.52	3.65	-2.64	6.95
206	1.2(D+Cl)+1.6Lr+0.8WL2	36.65	31.99	39.04	2.45	-0.25	4.63
207	1.2(D+Cl)+1.6Lr+0.8WR	39.57	28.70	45.15	4.21	-2.08	7.99
208	1.2(D+Cl)+1.6Lr+0.8WR	39.34	34.68	41.73	3.02	0.32	5.68
209	1.2(D+Cl)+1.6Lr+0.8W_L	39.93	29.06	45.51	4.29	-2.01	8.13
210	1.2(D+Cl)+1.6Lr+0.8W_L	39.69	35.03	42.08	3.09	0.40	5.82
211	1.2(D+Cl)+1.6Lr+0.8W_L	34.00	34.00	34.00	1.04	1.04	1.93
212	1.2(D+Cl)+1.6Lr+0.8W_L	36.86	36.86	36.86	1.64	1.64	3.05
213	1.2(D+Cl)+0.5Lr+1.6WL1	27.42	5.70	38.59	3.64	-8.94	7.09
214	1.2(D+Cl)+0.5Lr+1.6WR	26.86	17.54	31.64	1.24	-4.15	2.44
215	1.2(D+Cl)+0.5Lr+1.6WL2	32.69	10.96	43.86	4.76	-7.83	9.16
216	1.2(D+Cl)+0.5Lr+1.6WR	32.24	22.92	37.02	2.38	-3.02	4.55
217	1.2(D+Cl)+0.5Lr+1.6W_L	33.42	11.70	44.59	4.91	-7.67	9.44
218	1.2(D+Cl)+0.5Lr+1.6W_L	32.93	23.62	37.71	2.52	-2.87	4.82
219	1.2(D+Cl)+0.5Lr+1.6W_L	21.55	21.55	21.55	-1.59	-1.59	-2.97
220	1.2(D+Cl)+0.5Lr+1.6W_L	27.27	27.27	27.27	-0.39	-0.39	-0.72
221	1.2D+SL	34.31	34.31	34.31	1.10	1.10	2.05
222	1.2D+SR	34.83	34.83	34.83	1.21	1.21	2.25
223	0.9D+1.6WL1	24.18	2.46	35.35	2.96	-9.63	5.81
224	0.9D+1.6WL2	23.61	14.30	28.39	0.55	-4.84	1.16
225	0.9D+1.6WR1	29.44	7.72	40.61	4.07	-8.52	7.88
226	0.9D+1.6WR2	28.99	19.68	33.77	1.69	-3.70	3.27
227	0.9D+1.6W_Ln1	30.17	8.45	41.34	4.22	-8.36	8.17
228	0.9D+1.6W_Ln2	29.69	20.37	34.47	1.84	-3.56	3.55
229	0.9D+1.6W_Ln-1	18.31	18.31	18.31	-2.28	-2.28	-4.24
230	0.9D+1.6W_Ln-2	24.03	24.03	24.03	-1.07	-1.07	-1.99
231	0.9D+1.0SL	33.31	33.31	33.31	0.89	0.89	1.65
232	0.9D+1.0SR	33.83	33.83	33.83	1.00	1.00	1.86

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Sample calculation of BM & SF

Load case : 201

p_{1z} = 35.3-(35.3-35.3)*0.65/1.8= 35.30 kN/m^2

p_{2z} = 35.3+(35.3-35.3)*0.65/1.8= 35.30 kN/m^2 Overhang = 0.65 m

p_{3z} = 35.3-(35.3-35.3)*(0.65-0.257)/1.8= 35.30 kN/m^2 Effective depth = 0.257 m

BM_{1z} = (35.30-29.1)*0.65*0.65/2+0.5*(35.3-35.30)*0.65*0.65*0.67 = 1.31 kN-m/m

BM_{2z} = (35.30-29.1)*0.65*0.65/2-0.5*(-35.3+35.30)*0.65*0.65*0.67 = 1.31 kN-m/m

SF_{3z} = (35.30-29.1)*(0.65-0.257)+0.5*(35.3-35.30)*(0.65-0.257) = 2.44 kN/m

Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	4.91	0.074	0.099	0.0002	63.61	462.60	12ϕT	244.48	300	200	565.5
At top	-9.63	-0.146	-0.194	-0.0005	-1.19	462.60	12ϕT	244.48	300	200	565.5

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc) = 0.740 N/mm2

Shear at a distance 'd' from face of column, V_u= 6.95 kN/m

Shear stress , τ_v= (6.95*1000)/(1000*257)

= 0.03 N/mm2 **O.K.**

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :				
	At Bottom	12ϕT	200	c/c
	At Top	12ϕT	200	c/c
Along Z Direction :				
	At Bottom	12ϕT	200	c/c
	At Top	12ϕT	200	c/c

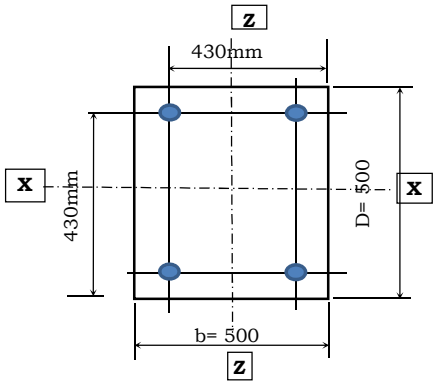
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5.4 Design calculation of column C2

COLUMN MKD C2



Input:

Clear cover to reinf.=	50mm	Table 20.6.1.3.1
Dia. of main reinf.=	20mm	
Dia. of link=	10mm	
Effective cover =	70mm	
Grade of Concrete (f_c') =	M35	
Grade of Steel (f_y) =	Fy420	
Grade of Reinf(links).=	Fy420	
Min. Steel Ratio	0.50%	
Max. Steel Ratio	4.00%	
Self weight of Column =	9.6kN	

For Compression-controlled sections

Strength reduction factor ϕ = 0.65 Table 21.2.2

Unsupproted length of column about X (l_{ux}) =	1600mm
Unsupproted length of column about Z (l_{uz}) =	1600mm

For Columns not braced against sidesway, Where, Effective length factor K= 2

Slenderness ratio,	$K\,l_{ux} / r_x$ =	21.33	< 22 SECTION 6.2.5a	Radius of gyration r_x=	150	mm
	$K\,l_{uz} / r_z$ =	21.33	< 22 SECTION 6.2.5a	Radius of gyration r_z=	150	mm

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γ_y = 0.76

γ_z = 0.76

Load Case	Load Combination	Factored Load From			Load at Column Base			Nominal Load/Moment			Kn = $P_n/f_c bD$ (N/mm^2)
		Pu (kN)	Hux (kN)	Huz (kN)	Pu (kN)	Muz (kNm)	Mux (kNm)	Pn (kN)	Mnz (kNm)	Mnx (kNm)	
201	1.4(D+Cl)	15.37	0.0	0.0	24.97	0.0	0.0	38.4	0.0	0.0	0.004
202	1.4(D+Cl+Cr)	24.25	0.0	0.0	33.85	0.0	0.0	52.1	0.0	0.0	0.006
203	1.4(D+Cl+T1)+0.5Lr	21.64	0.0	0.0	31.24	0.0	0.0	48.1	0.0	0.0	0.005
204	1.4(D+Cl+T2)+0.5Lr	23.29	0.0	0.0	32.89	0.0	0.0	50.6	0.0	0.0	0.006
205	1.2(D+Cl)+1.6Lr+0.8WL1	3.06	0.0	-9.5	12.66	0.0	15.2	19.5	0.0	23.4	0.002
206	1.2(D+Cl)+1.6Lr+0.8WL2	12.19	0.0	-4.1	21.79	0.0	6.5	33.5	0.0	10.0	0.004
207	1.2(D+Cl)+1.6Lr+0.8WR1	11.58	0.0	9.5	21.18	0.0	15.2	32.6	0.0	23.4	0.004
208	1.2(D+Cl)+1.6Lr+0.8WR2	20.90	0.0	4.1	30.50	0.0	6.5	46.9	0.0	10.0	0.005
209	1.2(D+Cl)+1.6Lr+0.8W_Ln1	12.77	0.0	-9.5	22.37	0.0	15.2	34.4	0.0	23.4	0.004
210	1.2(D+Cl)+1.6Lr+0.8W_Ln2	22.03	0.0	-4.1	31.63	0.0	6.5	48.7	0.0	10.0	0.006
211	1.2(D+Cl)+1.6Lr+0.8W_Ln-1	11.14	0.0	0.0	20.74	0.0	0.0	31.9	0.0	0.0	0.004
212	1.2(D+Cl)+1.6Lr+0.8W_Ln-2	20.41	0.0	0.0	30.01	0.0	0.0	46.2	0.0	0.0	0.005
213	1.2(D+Cl)+0.5Lr+1.6WL1	-45.35	0.0	-19.0	-35.75	0.0	30.4	-55.0	0.0	46.8	-0.006
214	1.2(D+Cl)+0.5Lr+1.6WR1	-27.08	0.0	-8.1	-17.48	0.0	13.0	-26.9	0.0	20.0	-0.003
215	1.2(D+Cl)+0.5Lr+1.6WL2	-28.29	0.0	19.0	-18.69	0.0	30.4	-28.8	0.0	46.8	-0.003
216	1.2(D+Cl)+0.5Lr+1.6WR2	-9.65	0.0	8.1	-0.05	0.0	13.0	-0.1	0.0	20.0	0.000
217	1.2(D+Cl)+0.5Lr+1.6W_Ln1	-25.93	0.0	-19.0	-16.33	0.0	30.4	-25.1	0.0	46.8	-0.003
218	1.2(D+Cl)+0.5Lr+1.6W_Ln2	-7.40	0.0	-8.1	2.20	0.0	13.0	3.4	0.0	20.0	0.000
219	1.2(D+Cl)+0.5Lr+1.6W_Ln-1	-29.17	0.0	0.0	-19.57	0.0	0.0	-30.1	0.0	0.0	-0.003
220	1.2(D+Cl)+0.5Lr+1.6W_Ln-2	-10.65	0.0	0.0	-1.05	0.0	0.0	-1.6	0.0	0.0	0.000
221	1.2D+SL	12.17	0.0	0.0	21.77	0.0	0.0	33.5	0.0	0.0	0.004
222	1.2D+SR	13.85	0.0	0.0	23.45	0.0	0.0	36.1	0.0	0.0	0.004
223	0.9D+1.6WL1	-55.86	0.0	-19.0	-46.26	0.0	30.4	-71.2	0.0	46.8	-0.008
224	0.9D+1.6WL2	-37.59	0.0	-8.1	-27.99	0.0	13.0	-43.1	0.0	20.0	-0.005
225	0.9D+1.6WR1	-38.80	0.0	19.0	-29.20	0.0	30.4	-44.9	0.0	46.8	-0.005
226	0.9D+1.6WR2	-20.16	0.0	8.1	-10.56	0.0	13.0	-16.3	0.0	20.0	-0.002
227	0.9D+1.6W_Ln1	-36.44	0.0	-19.0	-26.84	0.0	30.4	-41.3	0.0	46.8	-0.005
228	0.9D+1.6W_Ln2	-17.91	0.0	-8.1	-8.31	0.0	13.0	-12.8	0.0	20.0	-0.001
229	0.9D+1.6W_Ln-1	-39.68	0.0	0.0	-30.08	0.0	0.0	-46.3	0.0	0.0	-0.005
230	0.9D+1.6W_Ln-2	-21.16	0.0	0.0	-11.56	0.0	0.0	-17.8	0.0	0.0	-0.002
231	0.9D+1.0SL	8.92	0.0	0.0	18.52	0.0	0.0	28.5	0.0	0.0	0.003
232	0.9D+1.0SR	10.60	0.0	0.0	20.20	0.0	0.0	31.1	0.0	0.0	0.004

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Reinforcement: Provide,

4-Nos	20 mm Dia
2-Nos	20 mm Dia
2-Nos	20 mm Dia

1257.14 Sq mm (At Corner)
628.57 Sq mm (Along Z axis)
628.57 Sq mm (Along Y axis)

A_{sc} pvd. = 2514.29 Sq mm

Total A_{sc} = 1.01 % ρ_g = 0.010

β = P_n / P_0 (range is 0.55 to 0.70) Where, $P_0 = 0.85 f'_c (A_g - A_{sc}) + f_y A_{sc}$ = 841.87 T SECTION 22.4.2.2

Load Comb	From Chart of SP17(M)-R35-		M_{noz}	M_{nox}	β	$\alpha_n = \log(0.5)/\log(\beta)$	Interaction ratio	Remarks
	$R_{nz}=M_{noz}/f'_cb^2D$	$R_{nx}=M_{nox}/f'_cb^2D$						
1.4(D+Cl)	0.055	0.055	240.6	240.6	0.55	1.16	0.00	O.K
1.4(D+Cl+Cr)	0.055	0.055	240.6	240.6	0.55	1.16	0.00	O.K
1.4(D+Cl+T1)+0.5Lr	0.055	0.055	240.6	240.6	0.55	1.16	0.00	O.K
1.4(D+Cl+T2)+0.5Lr	0.055	0.055	240.6	240.6	0.55	1.16	0.00	O.K
1.2(D+Cl)+1.6Lr+0.8WL1	0.055	0.055	240.6	240.6	0.55	1.16	0.07	O.K
1.2(D+Cl)+1.6Lr+0.8WL2	0.055	0.055	240.6	240.6	0.55	1.16	0.03	O.K
1.2(D+Cl)+1.6Lr+0.8WR1	0.055	0.055	240.6	240.6	0.55	1.16	0.07	O.K
1.2(D+Cl)+1.6Lr+0.8WR2	0.055	0.055	240.6	240.6	0.55	1.16	0.03	O.K
1.2(D+Cl)+1.6Lr+0.8W_Ln1	0.055	0.055	240.6	240.6	0.55	1.16	0.07	O.K
1.2(D+Cl)+1.6Lr+0.8W_Ln2	0.055	0.055	240.6	240.6	0.55	1.16	0.03	O.K
1.2(D+Cl)+1.6Lr+0.8W_Ln-1	0.055	0.055	240.6	240.6	0.55	1.16	0.00	O.K
1.2(D+Cl)+1.6Lr+0.8W_Ln-2	0.055	0.055	240.6	240.6	0.55	1.16	0.00	O.K
1.2(D+Cl)+0.5Lr+1.6WL1	0.055	0.055	240.6	240.6	0.55	1.16	0.15	O.K
1.2(D+Cl)+0.5Lr+1.6WR1	0.055	0.055	240.6	240.6	0.55	1.16	0.06	O.K
1.2(D+Cl)+0.5Lr+1.6WL2	0.055	0.055	240.6	240.6	0.55	1.16	0.15	O.K
1.2(D+Cl)+0.5Lr+1.6WR2	0.055	0.055	240.6	240.6	0.55	1.16	0.06	O.K
1.2(D+Cl)+0.5Lr+1.6W_Ln1	0.055	0.055	240.6	240.6	0.55	1.16	0.15	O.K
1.2(D+Cl)+0.5Lr+1.6W_Ln2	0.055	0.055	240.6	240.6	0.55	1.16	0.06	O.K
1.2(D+Cl)+0.5Lr+1.6W_Ln-1	0.055	0.055	240.6	240.6	0.55	1.16	0.00	O.K
1.2(D+Cl)+0.5Lr+1.6W_Ln-2	0.055	0.055	240.6	240.6	0.55	1.16	0.00	O.K
1.2D+SL	0.055	0.055	240.6	240.6	0.55	1.16	0.00	O.K
1.2D+SR	0.055	0.055	240.6	240.6	0.55	1.16	0.00	O.K
0.9D+1.6WL1	0.055	0.055	240.6	240.6	0.55	1.16	0.15	O.K
0.9D+1.6WL2	0.055	0.055	240.6	240.6	0.55	1.16	0.06	O.K
0.9D+1.6WR1	0.055	0.055	240.6	240.6	0.55	1.16	0.15	O.K
0.9D+1.6WR2	0.055	0.055	240.6	240.6	0.55	1.16	0.06	O.K
0.9D+1.6W_Ln1	0.055	0.055	240.6	240.6	0.55	1.16	0.15	O.K
0.9D+1.6W_Ln2	0.055	0.055	240.6	240.6	0.55	1.16	0.06	O.K
0.9D+1.6W_Ln-1	0.055	0.055	240.6	240.6	0.55	1.16	0.00	O.K
0.9D+1.6W_Ln-2	0.055	0.055	240.6	240.6	0.55	1.16	0.00	O.K
0.9D+1.0SL	0.055	0.055	240.6	240.6	0.55	1.16	0.00	O.K
0.9D+1.0SR	0.055	0.055	240.6	240.6	0.55	1.16	0.00	O.K

Links, 10 mm dia @ 250 C/C

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REFERENCES

Reference is made to codes, standards and specifications as indicated in Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST- CRT-1001) Clause 4.1

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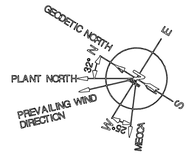
ATTACHMENT 1 OTE INPUT DRAWING & LOAD DATA DRAWING

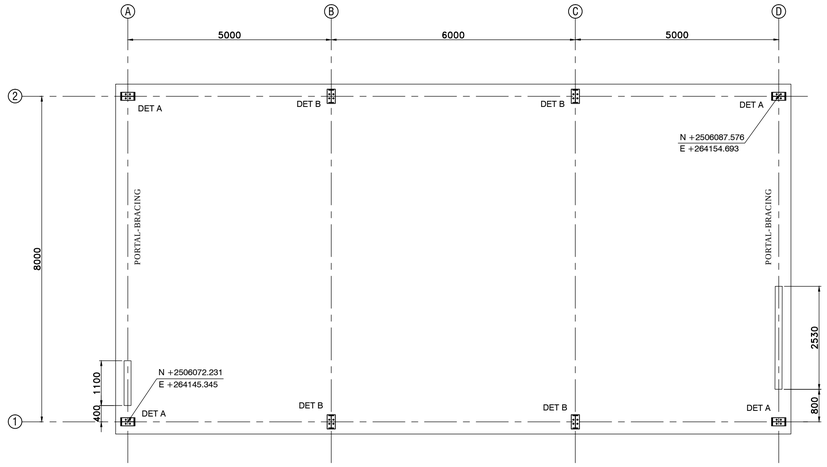
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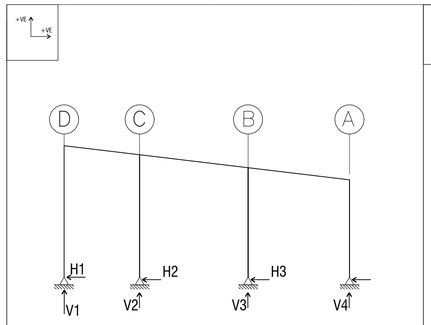
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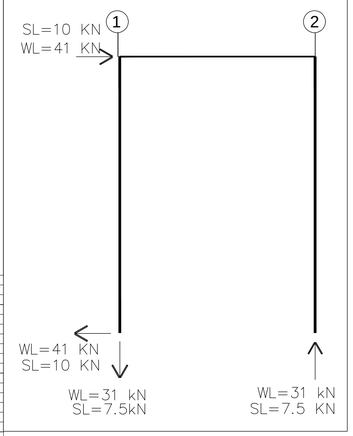
ANCHOR BOLT PLAN

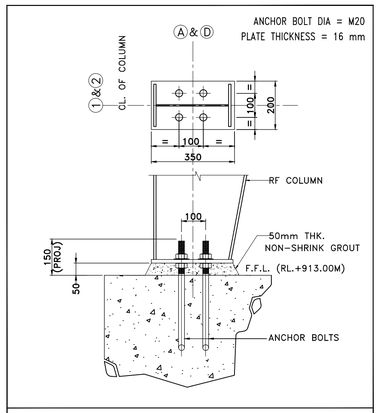


TYPICAL RIGID FRAME

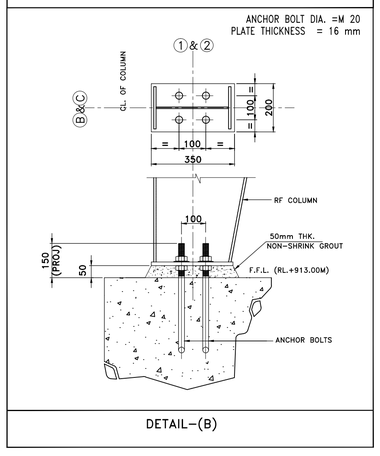
CATEGORY	LOAD CASE	LEFT COL.		INT. COL.1			INT. COL.2			RIGHT COL.	
		H1 (KN)	V1 (KN)	H2 (KN)	V2 (KN)	Z2 (KN)	H3 (KN)	V3 (KN)	Z3 (KN)	H4 (KN)	V4 (KN)
Seismic	Seismic_L	-0.40	-0.54	0.00	0.70	0.00	0.00	-0.84	0.00	-0.60	0.67
	Seismic_R	0.40	0.54	0.00	-0.70	0.00	0.00	0.84	0.00	0.60	-0.67
Dead	Dead	0.40	8.22	0.00	10.53	0.00	0.00	10.84	0.00	-0.40	7.91
Collateral	Collateral	1.78	10.36	0.00	-0.30	0.00	0.00	0.14	0.00	-1.78	10.20
Roof Live	Live	0.44	5.17	0.00	14.00	0.00	0.00	14.18	0.00	-0.44	5.11
CRANE	CRANE	-0.16	-1.05	0.00	5.92	0.00	0.00	6.34	0.00	0.16	-1.22
Wind	Wind_Left1	-15.26	-24.87	0.00	-26.26	-12.55	0.00	-41.01	-11.88	-18.55	4.56
	Wind_Left2	-19.21	-19.46	0.00	-16.14	-5.38	0.00	-29.59	-5.09	-14.69	9.52
	Wind_Right1	11.99	-3.51	0.00	-31.08	12.55	0.00	-30.35	11.88	13.57	-22.42
	Wind_Right2	8.08	1.97	0.00	-21.14	5.38	0.00	-18.70	5.09	17.52	-17.57
	Wind_Long1	5.89	-17.13	0.00	-25.88	-12.55	0.00	-28.87	-11.88	-7.28	-10.67
	Wind_Long2	1.96	-11.66	0.00	-15.83	-5.38	0.00	-17.29	-5.09	-3.36	-5.77
	Wind_Long-1	5.92	-13.37	0.00	-23.94	0.00	0.00	-30.90	0.00	-7.31	-14.36
	Wind_Long-2	2.00	-7.91	0.00	-13.88	0.00	0.00	-19.32	0.00	-3.40	-9.45
Temperature	25Degree	0.66	0.45	0.00	-0.39	0.00	0.00	-0.59	0.00	-0.65	0.54
	-25Degree	-0.65	-0.45	0.00	0.39	0.00	0.00	0.59	0.00	0.65	-0.54

SIDEWALL PORTAL BRACING REACTION AT GL-A & D





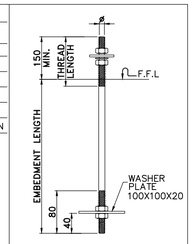
DETAIL–(A)



DETAIL–(B)

NOTE: ANCHOR BOLT SPECIFICATION F1554(GRADE36,55or105)

QTY.	DIA. OF BOLT	DIA. OF HOLE	THREAD LENGTH	ANCHOR ROD LENGTH
	ø	D	TL	RL
32	M20	26	150	450
	M24	30	150	500
	M30	36	200	700
	M36	42	200	1000
	>M36	ø + 8	200	AS PER DESIGN



ANCHOR BOLT DETAIL

MATERIAL SPECIFICATIONS

NO.	MATERIALS		GRADE & SPECIFICATION	STRENGTH
01	ANCHOR BOLTS	J-type	ASTM F1554 Grade A36M, Hot dip galvanized to ASTM A153M, Heavy Hex Nuts A563M-10S Washers-ASTMF436	Fu = 400 N/mm^2
		STRAIGHT -type	BS EN ISO 898-1 GRDE 8.8, Hot dip galvanized to BS EN ISO1461 BS EN ISO 898-2 10 STYLE2 WASHER-BS4320	Fu = 800 N/mm^2
02	BUILT-UP SECTIONS		ASTM A572M Grade 50 or equivalent	Fy = 345 N/mm^2
03	HOT ROLLED SECTIONS	Tubes	E10025 S355HR or equivalent	Fy = 345 N/mm^2
		I-Sections	ASTM A572M Grade 50 or equivalent	Fy = 345 N/mm^2
		Channels	E10025 S355JR or equivalent	Fy = 355 N/mm^2
		Angles	ASTM A572M Grade 50 or equivalent	Fy = 345 N/mm^2
		Pipes	ASTM A53 Grade B or equivalent	Fy = 240 N/mm^2
04	Primary Connection bolts		ASTM A325M TYPE-1; Fully threaded hot dip galvanized	Ft = 827 N/mm^2
05	DIAGONAL "X" BRACING	Cables	ASTM A475-Extra High Strength Class A, HDG	BREAKING LOAD pu= 119.7 kN
		HR Angles	ASTM A572M Grade 50 or equivalent	Fy = 345 N/mm^2
06	Cold formed secondary members		ASTM A653M SS Grade 340 Class 1: Coating Z275 (Grade 90)	Fy = 344 N/mm^2
07	Secondary Connection bolts (for purlins,girts and flange braces)		Grade 8.8 Fully threaded hot dip galvanized	Ft = 400 N/mm^2
08	Sag Rods		ASTM Grade A36M Electro galvanized, Fully threaded with 4 nuts Coating Z275 (Grade 90)	Fy = 245 N/mm^2
09	Mezzanine Deck Panels		ASTM A653M SS Grade 340 Class 1: Coating Z275 (Grade 90)	Fy = 344 N/mm^2

LOADS LEGEND	
Wind_Left1	Denotes wind load from left to right with internal pressure
Wind_Right1	Denotes wind load from right to left with internal SUCTION
Wind_Left2	Denotes wind load from left to right with internal pressure
Wind_Right2	Denotes wind load from right to left with internal SUCTION
Wind_Long1	Denotes wind load parallel to ridge from end wall with internal pressure
Wind_Long2	Denotes wind load parallel to ridge from end wall with internal suction
Wind_Long–1	Denotes wind load parallel to ridge from end wall with internal pressure
Wind_Long–2	Denotes wind load parallel to ridge from end wall with internal suction
Seismic_R	Denotes seismic load from left
Seismic_L	Denotes seismic load from right

IMPORTANT NOTES :
(1) BEARING CAPACITY OF CONCRETE UNDER BASE PLATE SHOULD NOT BE LESS THAN 1.05 KN/CM2
(2) GIVEN REACTIONS ARE FOR UNFACTORED LOADS.
(3) BRACING REACTIONS TO BE ADDED WITH MAIN FRAME COLUMN REACTIONS FOR THE DESIGN OF FOUNDATION.
(4) LOADS IN THE SAME CATEGORY , NOT TO BE COMBINED IN THE SAME LOAD COMBINATION .

NOTES–
1. ALL DIMENSIONS ARE IN MILLIMETER.
2. ALL LEVELS AND CO–ORDINATE ARE IN METER
3. F.F.L. = FINISH FLOOR LEVEL.

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3840-EG-ME-GAD-1002-A01	LOAD PLAN-REAGENTS FLOCCULANT-PLAN VIEW								
MD-216-3840-EG-ME-GAD-1001-A01	LOAD PLAN-REAGENTS FLOCCULANT-PLAN VIEW	A01	20/06/2020	ISSUED FOR REFERENCE	---	SYC	---	PRI	

aic arabian international company
P.O.Box : 41017 Jeddah 21521,K.S.A
Tel. : +966 2 591 0999
Fax : +966 2 591 0099
E-mail : aic@aicsteel.com
Site : www.aicsteel.com

DRAWING NUMBER						REVISION
MD-P20131-B08-E001						00
DATE	17/06/2020	DATE	17/06/2020	DATE	17/06/2020	
DRAWN	7014-7017	CHECKED	7013	APPROVED	7001	

LARSEN & TOUBRO DRAWING NUMBER	REVISION
019045_450_S_G_26073_001	0
OUTOTEC DRAWING NUMBER	DOCUMENT ID / REVISION
900707_GAU01_AA02_MTL01_38405	OU100676484 / 0

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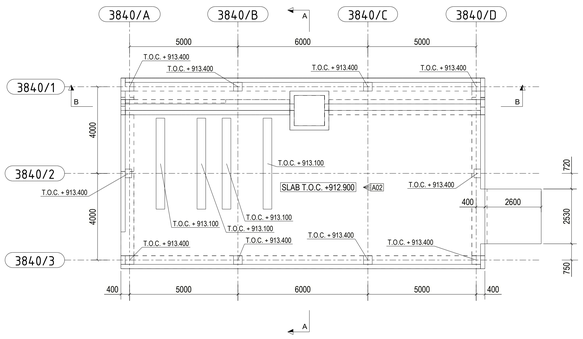
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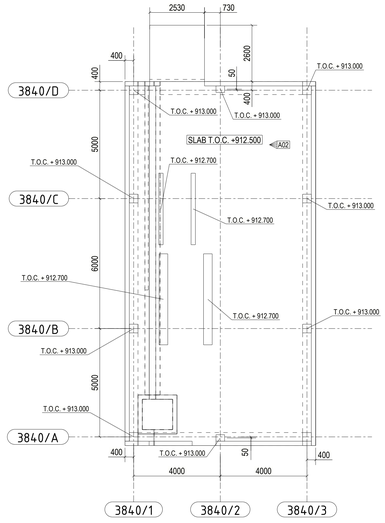
DRAWN	--	--	MANSOURAH - MASSARAH GOLD PROJECT
CHECKED	SYC	20/06/2020	GENERAL ARRANGEMENT DRAWING
ENGINEER	--	--	REAGENT-FLOCCULANT 1 - ACIDULATION
APPROVED	PRI	20/06/2020	ANCHOR BOLT PLAN & REACTIONS
EMPLOYER			
SCALE	NTS	A1	DRAWING NUMBER: MD-216-3840-EG-ST-GAD-2352 REV A01

DRAWING NUMBER: MD-216-3840-EG-CV-GAD-1001
REV: A02

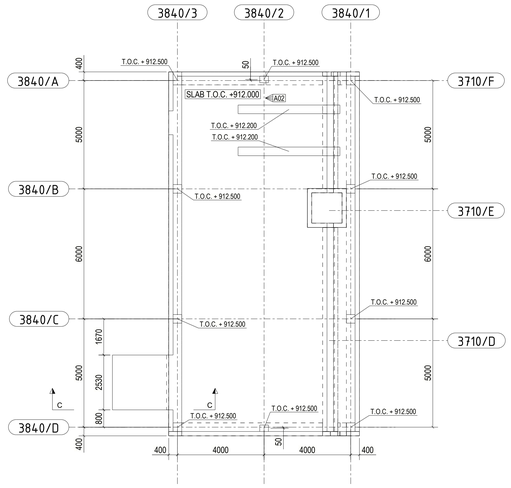
DET 1
1:100
TYPICAL DIMENSIONS (EXCEPT LEVELS)



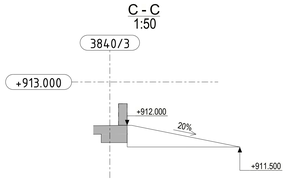
DET 3
1:100



DET 4
1:100

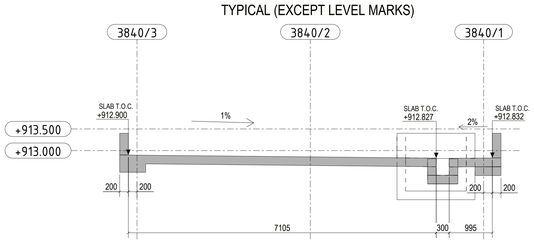


C - C
1:50

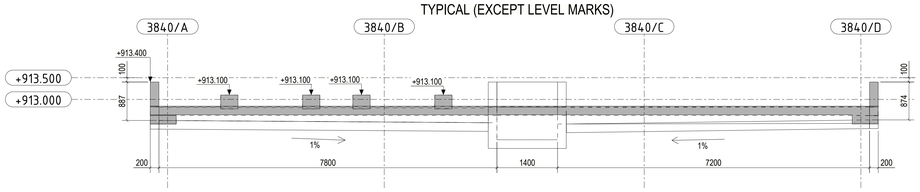


DET 2
1:100

A - A
1:50
TYPICAL (EXCEPT LEVEL MARKS)



B - B
1:50
TYPICAL (EXCEPT LEVEL MARKS)



PLAN VIEW
1:900

THIS DESIGN IS NOT ISSUED FOR CONSTRUCTION

A02 ① FINISHED TOP GROUND LEVEL (F.G.L) FOR PROCESS AREA 3840 = RL(+)912.900 m

A02 ② FINISHED TOP GROUND LEVEL (F.G.L) FOR PROCESS AREA 3840 = RL(+)912.000 m

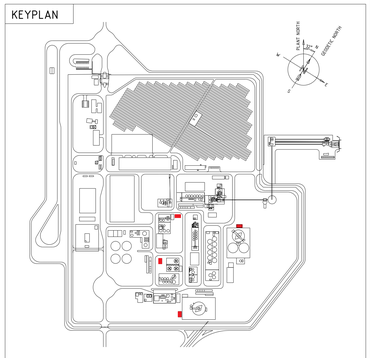
A02 ③ FINISHED TOP GROUND LEVEL (F.G.L) FOR PROCESS AREA 3840 = RL(+)912.500 m

A02 ④ FINISHED TOP GROUND LEVEL (F.G.L) FOR PROCESS AREA 3840 = RL(+)912.000 m

CONCRETE ELEMENT FLOOR (PRE-CAST SLAB)
CONCRETE FLOOR
CONCRETE FLOOR WITH ACID RESISTANT COATING
CONCRETE WALL

1. ALL DIMENSIONS ARE IN MILLIMETER, ELEVATIONS ARE IN METER UNLESS OTHERWISE STATED
2. CO-ORDINATES ARE SHOWN IN GENERAL ARRENGEMENT DRAWINGS
3. GRATING TYPE (FRP OR NORMAL) IS SPECIFIED IN GENERAL ARRENGEMENT DRAWINGS
4. F.F.L, F.G.L, T.O.C = FINISHED FLOOR LEVEL, FINISHED GROUND LEVEL, TOP OF CONCRETE, T.O.S, B.O.S, B.O.B, = TOP OF STEEL, BOTTOM OF STEEL, BOTTOM OF BASE PLATE
5. BUILDING DIMENSIONS AND HEIGHTS ARE TENTATIVE, SHALL BE FINALIZED DURING DETAIL ENGINEERING SUBJECT TO MINIMUM AS SHOWN.
6. ALL MAINTENANCE FACILITIES, HANDLING PROVISIONS, ACCESS WAYS, PLATFORMS FOR MAINTENANCE SHALL BE PROVIDED DURING DETAIL ENGINEERING WHICH ARE NOT SHOWN ENTIRELY. THIS DRAWING IS TO BE CONSIDERED AS CONCEPTUAL ARRANGEMENT OF THE FACILITIES.

KEYPLAN



DOCUMENT NUMBER	REFERENCE DOCUMENTS	DOCUMENT NUMBER	REFERENCE DOCUMENTS
MD-216-3840-EG-ST-GAD-1002	PLAN VIEW		
MD-216-1000-EG-ST-CRT-1001	CIVIL,STRUCTURAL AND ARCHITECTURAL DESIGN BASIS	MD-216-3840-EG-ST-GAD-1001	3D VIEW
MD-216-3840-EG-ME-GAD-1008	PLAN VIEW	MD-216-3840-EG-ME-GAD-1009	PLAN VIEW
MD-216-3840-EG-ME-GAD-1006	PLAN VIEW	MD-216-3840-EG-ME-GAD-1007	PLAN VIEW

REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
A02	22.04.2020	ISSUED FOR DETAIL DESIGN, LOCATION OF BUILDINGS 1, 3 AND 4, AND GRDS CHANGED	Kirish	MsLai	LuiPiz	SusCha	
A01	23.12.2019	ISSUED FOR APPROVAL	AnnFor	MsLai	LuiPiz	SusCha	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

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LARSEN & TOUBRO DRAWING NUMBER: REVISION 1

OUTOTEC DRAWING NUMBER: 900707_GAU01_AA02_CTA04_38401
DOCUMENT ID: OU100668158
REVISION: 1

DRAWN	AnnFor	23.12.2019
CHECKED	MsLai	23.12.2019
ENGINEER	LuiPiz	23.12.2019
APPROVED	SusCha	23.12.2019
EMPLOYER		
SCALE	1:100	A0

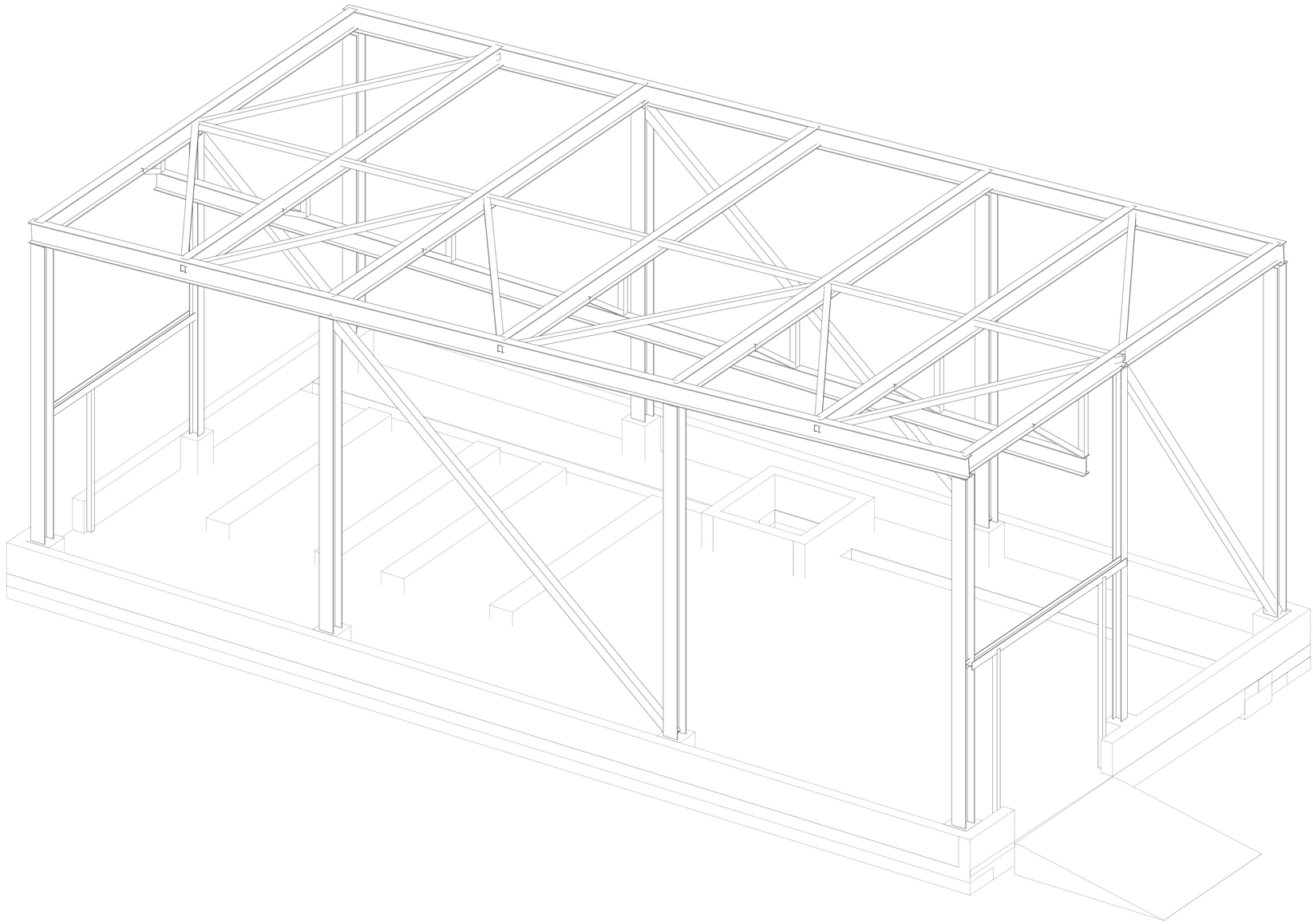
MANSOURAH - MASSARAH GOLD PROJECT
CIVIL WORKS GUIDE DRAWING
3840 FLOCCULANT
PLAN VIEW

DRAWING NUMBER: MD-216-3840-EG-CV-GAD-1001
REV: A02

Tekla Structures

DRAWING NUMBER: MD-216-3840-EG-ST-GAD-1001
REV: A02

3D VIEW - TYPICAL STRUCTURE IN AREA 3840



THIS DESIGN IS NOT ISSUED FOR CONSTRUCTION

A02 ▶ (1) FINISHED TOP GROUND LEVEL (F.G.L) FOR PROCESS AREA 3840 = RL(+)912.900 m

A02 ▶ (2) FINISHED TOP GROUND LEVEL (F.G.L) FOR PROCESS AREA 3840 = RL(+)912.000 m

A02 ▶ (3) FINISHED TOP GROUND LEVEL (F.G.L) FOR PROCESS AREA 3840 = RL(+)912.500 m

A02 ▶ (4) FINISHED TOP GROUND LEVEL (F.G.L) FOR PROCESS AREA 3840 = RL(+)912.000 m

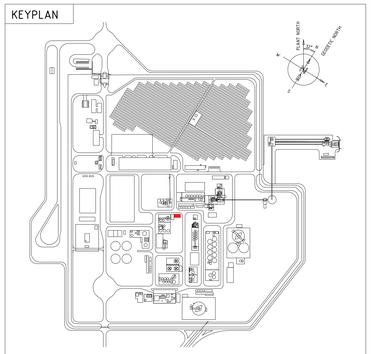
CHECKERED PLATE

GRATING

GRATING - FRP

1. ALL DIMENSIONS ARE IN MILLIMETER, ELEVATIONS ARE IN METER UNLESS OTHERWISE STATED
2. CO-ORDINATES ARE SHOWN IN GENERAL ARRENGEMENT DRAWINGS
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5. BUILDING DIMENSIONS AND HEIGHTS ARE TENTATIVE, SHALL BE FINALIZED DURING DETAIL ENGINEERING SUBJECT TO MINIMUM AS SHOWN.
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KEYPLAN



DOCUMENT NUMBER	REFERENCE DOCUMENTS
MD-216-3840-EG-ST-GAD-1002	PLAN VIEW
MD-216-1000-EG-ST-CRT-1001	CIVIL,STRUCTURAL AND ARCHITECTURAL DESIGN BASIS
MD-216-3844-EG-ME-GAD-1008	PLAN VIEW
MD-216-3845-EG-ME-GAD-1006	PLAN VIEW

DOCUMENT NUMBER	REFERENCE DOCUMENTS
MD-216-3840-EG-CV-GAD-1001	PLAN VIEW
MD-216-3840-EG-ME-GAD-1009	PLAN VIEW
MD-216-3840-EG-ME-GAD-1003	PLAN VIEW

REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
A02	22.04.2020	ISSUED FOR DETAIL DESIGN, LOCATION OF BUILDINGS 1, 3 AND 4 AND GRIDS CHANGED	KirJak	MikLai	LuiPiz	SueCha	
A01	23.12.2019	ISSUED FOR APPROVAL	AnnFor	MikLai	LuiPiz	SueCha	

REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP

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LARSEN & TOUBRO DRAWING NUMBER: 900707_GAU01_AA02_CTA05_38401
REVISION: 1
OUTOTEC DRAWING NUMBER: OU100668152
DOCUMENT ID
REVISION: 1

DRAWN	AnnFor	23.12.2019
CHECKED	MikLai	23.12.2019
ENGINEER	LuiPiz	23.12.2019
APPROVED	SueCha	23.12.2019
EMPLOYER		
SCALE	1:25	A0

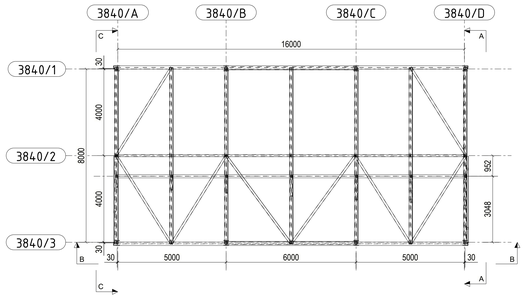
MANSOURAH - MASSARAH GOLD PROJECT
STEEL STRUCTURE GUIDE DRAWING
3840 FLOCCULANT
3D VIEW

DRAWING NUMBER: MD-216-3840-EG-ST-GAD-1001
REV: A02

Tekla Structures

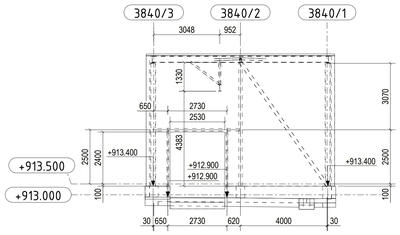
DET 1
1:100

TYPICAL DIMENSIONS



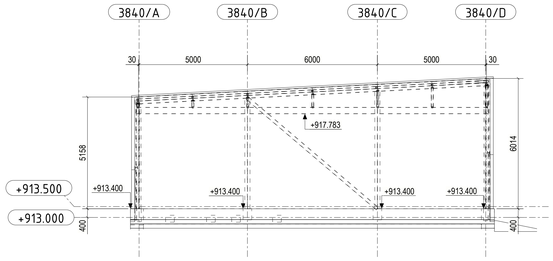
A - A
1:100

TYPICAL MEASURES (EXPECT LEVELS)



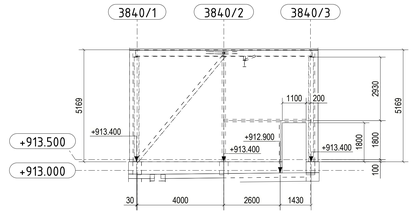
B - B
1:100

TYPICAL MEASURES (EXPECT LEVELS)

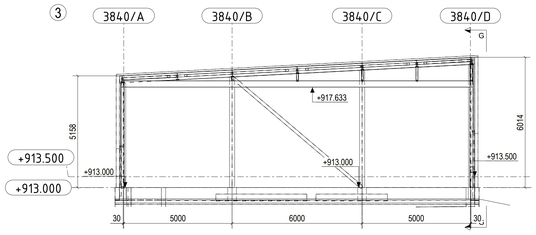


C - C
1:100

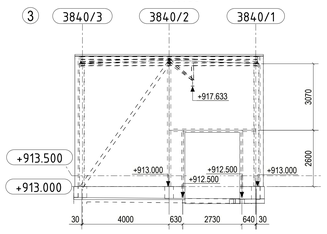
TYPICAL MEASURES (EXPECT LEVELS)



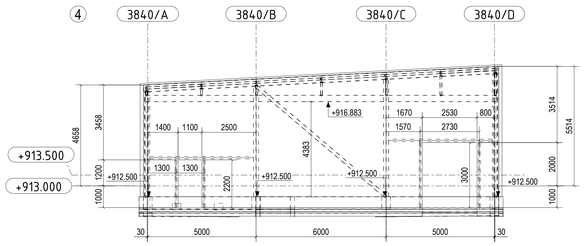
D - D
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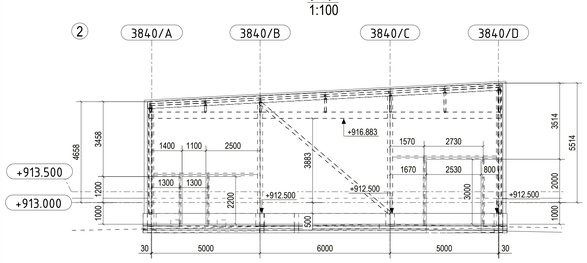
G - G
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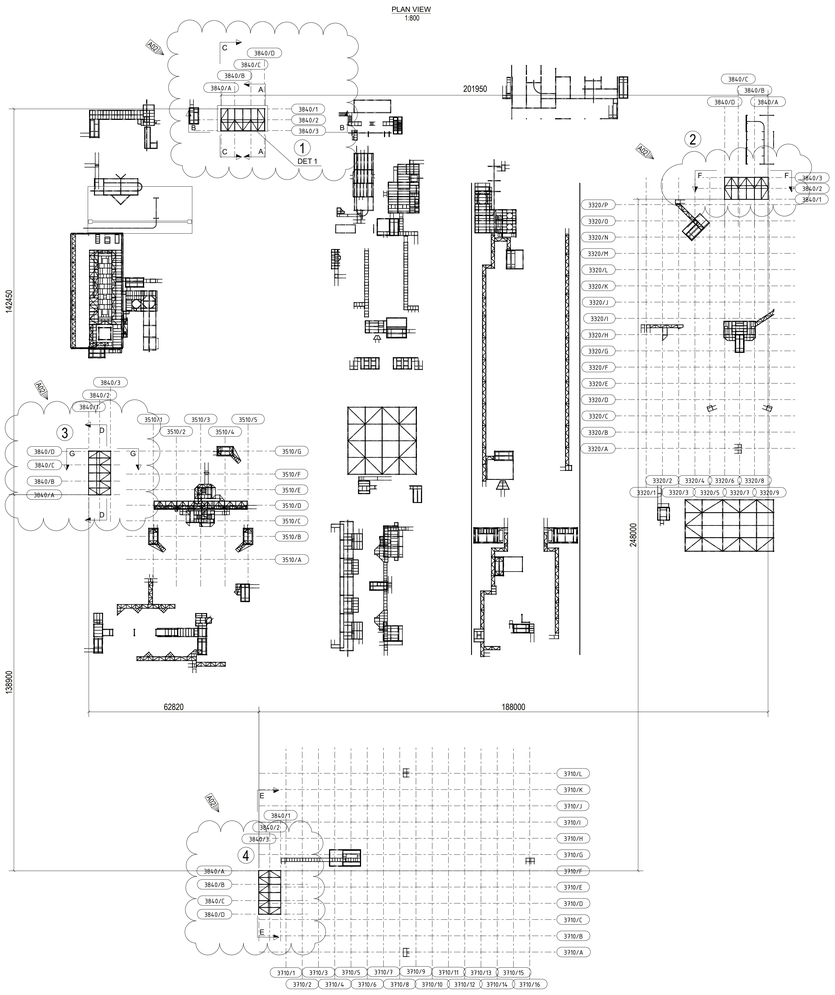
E - E
1:100



F - F
1:100



PLAN VIEW
1:800



THIS DESIGN IS NOT ISSUED FOR CONSTRUCTION

A02 ① FINISHED TOP GROUND LEVEL (F.G.L) FOR PROCESS AREA 3840 = RL(+)912.900 m

A02 ② FINISHED TOP GROUND LEVEL (F.G.L) FOR PROCESS AREA 3840 = RL(+)912.000 m

A02 ③ FINISHED TOP GROUND LEVEL (F.G.L) FOR PROCESS AREA 3840 = RL(+)912.500 m

A02 ④ FINISHED TOP GROUND LEVEL (F.G.L) FOR PROCESS AREA 3840 = RL(+)912.000 m

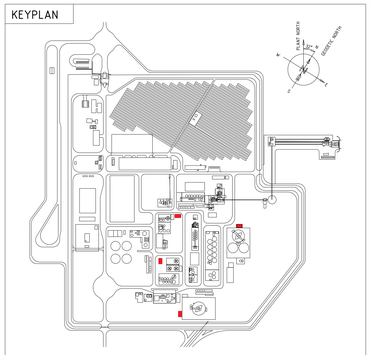
CHECKERED PLATE

GRATING

GRATING - FRP

1. ALL DIMENSIONS ARE IN MILLIMETER, ELEVATIONS ARE IN METER UNLESS OTHERWISE STATED
2. CO-ORDINATES ARE SHOWN IN GENERAL ARRENGEMENT DRAWINGS
3. GRATING TYPE (FRP OR NORMAL) IS SPECIFIED IN GENERAL ARRENGEMENT DRAWINGS
4. F.F.L, F.G.L, T.O.C = FINISHED FLOOR LEVEL, FINISHED GROUND LEVEL, TOP OF CONCRETE, T.O.S, B.O.S, B.O.B, = TOP OF STEEL, BOTTOM OF STEEL, BOTTOM OF BASE PLATE
5. BUILDING DIMENSIONS AND HEIGHTS ARE TENTATIVE, SHALL BE FINALIZED DURING DETAIL ENGINEERING SUBJECT TO MINMUM AS SHOWN.
6. ALL MAINTENANCE FACILITIES, HANDLING PROVISIONS, ACCESS WAYS, PLATFORMS FOR MAINTENANCE SHALL BE PROVIDED DURING DETAIL ENGINEERING WHICH ARE NOT SHOWN ENTIRELY. THIS DRAWING IS TO BE CONSIDERED AS CONCEPTUAL ARRANGEMENT OF THE FACILITIES.

KEYPLAN



DOCUMENT NUMBER	REFERENCE DOCUMENTS	DOCUMENT NUMBER	REFERENCE DOCUMENTS
MD-216-3840-EG-ST-GAD-1001	3D VIEW		
MD-216-1000-EG-ST-CRT-1001	CIVIL,STRUCTURAL AND ARCHITECTURAL DESIGN BASIS	MD-216-3840-EG-CV-GAD-1001	PLAN VIEW
MD-216-3840-EG-ME-GAD-1008	PLAN VIEW	MD-216-3840-EG-ME-GAD-1009	PLAN VIEW
MD-216-3840-EG-ME-GAD-1006	PLAN VIEW	MD-216-3840-EG-ME-GAD-1007	PLAN VIEW

REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
A02	22.04.2020	ISSUED FOR DETAIL DESIGN, LOCATION OF BUILDINGS 1, 3 AND 4 AND GRIDS CHANGED	Kir.Isk	Msk.Lai	LuPiz	SuoCha	
A01	23.12.2019	ISSUED FOR APPROVAL	AnnFor	Msk.Lai	LuPiz	SuoCha	

REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

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LARSEN & TOUBRO DRAWING NUMBER			REVISION 1
OUTOTEC DRAWING NUMBER 900707_GAU01_AA02_CTA05_38402		DOCUMENT ID OU100668153	REVISION 1
DRAWN	AnnFor	23.12.2019	
CHECKED	Msk.Lai	23.12.2019	
ENGINEER	LuPiz	23.12.2019	
APPROVED	SuoCha	23.12.2019	
EMPLOYER			
SCALE 1:100		A0	

MANSOURAH - MASSARAH GOLD PROJECT
STEEL STRUCTURE GUIDE DRAWING
3840 FLOCCULANT
PLAN VIEW

DRAWING NUMBER MD-216-3840-EG-ST-GAD-1002 REV A02

Tekla Structures

C01

ATTACHMENT 2 COMMENT RESOLUTION LOG SHEET

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Mansourah Massarah Gold Project

Comment Resolution Log Sheet / Compliance Report

PROJECT:	**MANSOURAH MASSARAH GOLD PROJECT**	**PROJECT NO.:**	**216**
STAGE:	**4 - PROJECT EXECUTION.**	**REVIEW LOG DOC. NO. & REV.:**	
ACONEX REF. NO. & DATE:	BECHTEL-WTRAN-003880, 25 July 2020		

STATUS CODE: A - APPROVED, C - APPROVED WITH COMMENTS, R - REVISE & RESUBMIT & O - OTHERS (INFORMATION, REFERENCE, REVIEWED)

DOCUMENT NUMBER	Area / Title	Comment No.	PMC/PMT COMMENT DATE: 25 July 2020	Comment by	EPC Response DATE: 27 July 2020	Status	Remarks
			General Comments (cc)				
MD-216-3840-EG-CV-CAL-1427-A01	3840	1	Revise Document Title 3 as "Flocculant Concrete" instead of Flocculant.	Mr. Doron Golan - Bechtel	Revised.	Closed	